ELECTRIC ENGINES

POWERTRAIN CONTROLLER FOR DIY ELECTRIC VEHICLES

LEAF ECU

Installation manual

Installation and setup guide for the powertrain controller built around Nissan Leaf components for DIY electric vehicles.



Revision 2.0 · October 2026
Interactive version with circuit highlighting:
https://diag.electricengines.com.ua/manual/en/

Contents

1 General Information 2
1.1 Controller Connectors 2
1.2 CAN Bus 2
1.3 How to Read the Diagrams 3
2 12 V Power System 4
3 Inverter 6
3.1 Leaf Inverter Gen 1 (ZE0) 6
3.2 Leaf Inverter Gen 2 (AZE0) 6
3.3 Leaf Inverter Gen 3 (ZE1) 7
4 PDM Module 9
5 Controls 10
5.1 Selector Buttons 10
5.2 Brake Vacuum Booster 10
6 Power Relays (Contactors) 12
7 Accelerator Pedal 14
8 BMS 16
8.1 Chevrolet Volt Gen 1 16
8.2 Nissan Leaf (ZE0 / AZE0) 20
8.3 Orion 2 32
9 Current Sensor 33
10 Cooling System 35
11 Brake Light (Regeneration) 37
12 CHAdeMO 38
13 Display 40
14 Selector 41
14.1 Leaf Selector 42
14.2 Momentary Buttons 43
15 OBD2 Diagnostics 44
15.1 OBD2 Connector 44
15.2 Software 44
16 Initial Setup 45
16.1 Start-up Procedure 45
16.2 Accelerator Pedal Setup 46
16.3 Controller Configuration 46
17 FAQ / Troubleshooting 48
A Complete ECU Pinout 50
B CAN Buses 52
Revision History 53

1 General Information

Leaf ECU is the master powertrain controller: it drives the inverter and the traction-battery contactors, handles charging through the PDM and CHAdeMO, controls cooling and peripherals, and exchanges data with the BMS, the display and the gear selector over CAN.



Fig. 1 Leaf ECU controller

1.1 Controller Connectors

The controller has two 24-way connectors:

- **X1** — black;
- **X2** — grey.

Pin numbering is moulded on the connectors: rows A–C and positions 1–8. Pins can be inserted into the mating connector from the back of the housing without taking it apart — just pull out the yellow lock at the front of the connector first.

In the text pins are written as X1-B2 — hover such a tag to see where the pin sits on the connector. The full pinout is in Appendix A.

1.2 CAN Bus

> **WARNING**
> All CAN bus wires must be twisted into twisted pairs!

The controller has four CAN buses; the main one is CAN0 Powertrain (X1-A3 / X1-B3), which connects the inverter, the PDM, the OBD2 socket, the display and the selector. See Appendix B for all buses.

1.3 How to Read the Diagrams

All diagrams in this manual share one style. A wire's colour shows its purpose, not the insulation colour:

COLOUR	PURPOSE
	Traction battery power cables (HV)
	+12 V permanent — from the battery
	+12 V switched — from the ignition switch or through a relay
	+5 V — sensor supply from the controller
	Ground
/	CAN-H / CAN-L
	Signals into the controller (sensors, buttons, pilot lines)
	Controller outputs (relay coils, LEDs)

Relay contacts are numbered **1–2**, the coil **4–3**: pin 4 receives the drive signal from the controller, pin 3 is ground.

In the web edition of this manual the diagrams are interactive: hover or tap a wire to highlight the whole circuit, with a list of its connections and a view of the connector with the relevant pin shown below. The Expand button opens a diagram full-screen with zoom.

2 12 V Power System

Version 4 controller boards have a new power supply circuit. The controller needs:

- **permanent +12 V** from the battery;
- separately, **+12 V from the ignition switch** (IGN KEY).

After the ignition key is turned off the controller stays on until the motor stops, then it opens the battery contactors and switches itself off.

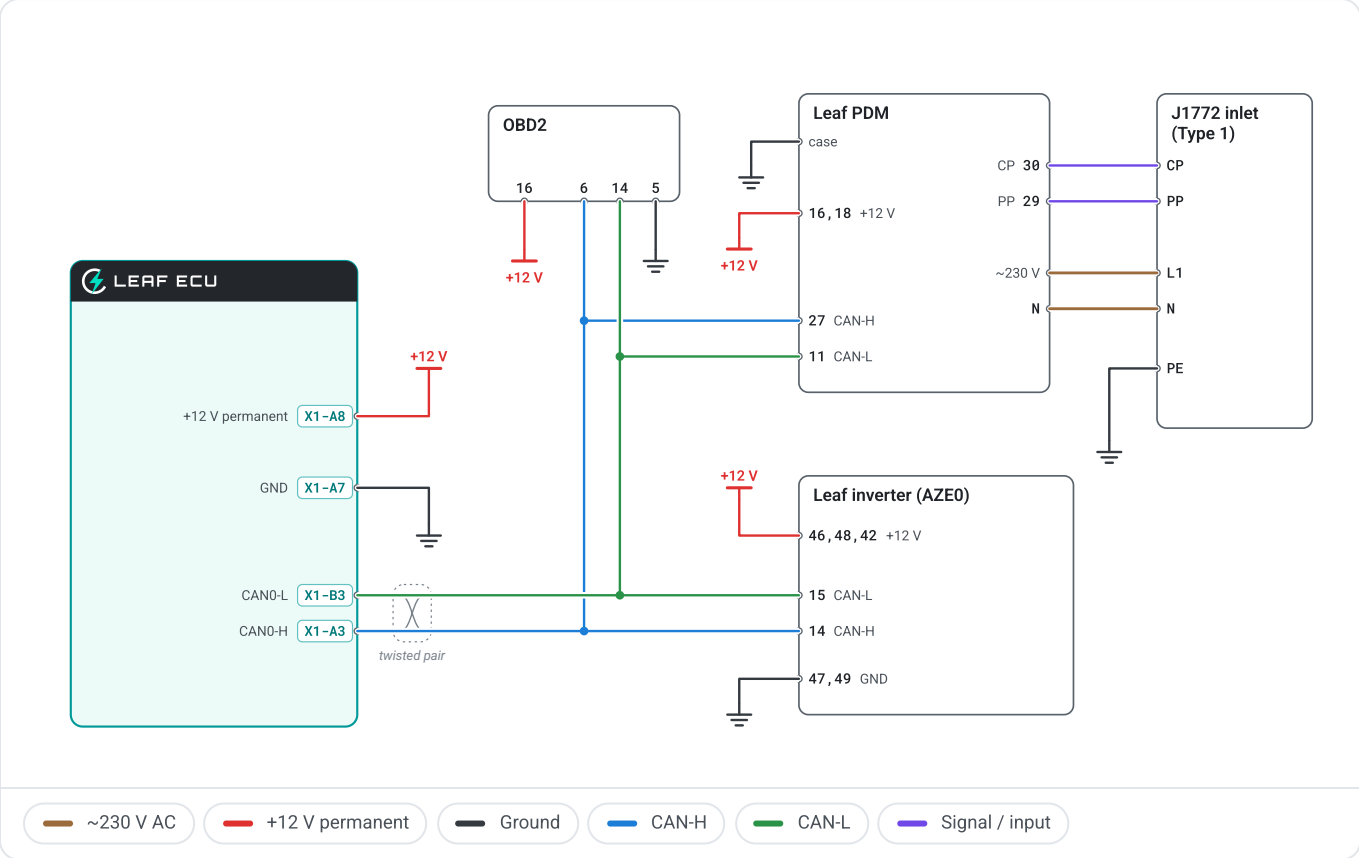


Fig. 2 Basic CAN bus and 12 V connection (for the AZE0 inverter; for other inverters see the connector pinouts below — the wiring is similar)

> **NOTE**
>
> The Leaf Type 1 charge inlet has resistor R5 right at the socket, under the rubber boot. It must be relocated and wired as shown in Fig. 3.

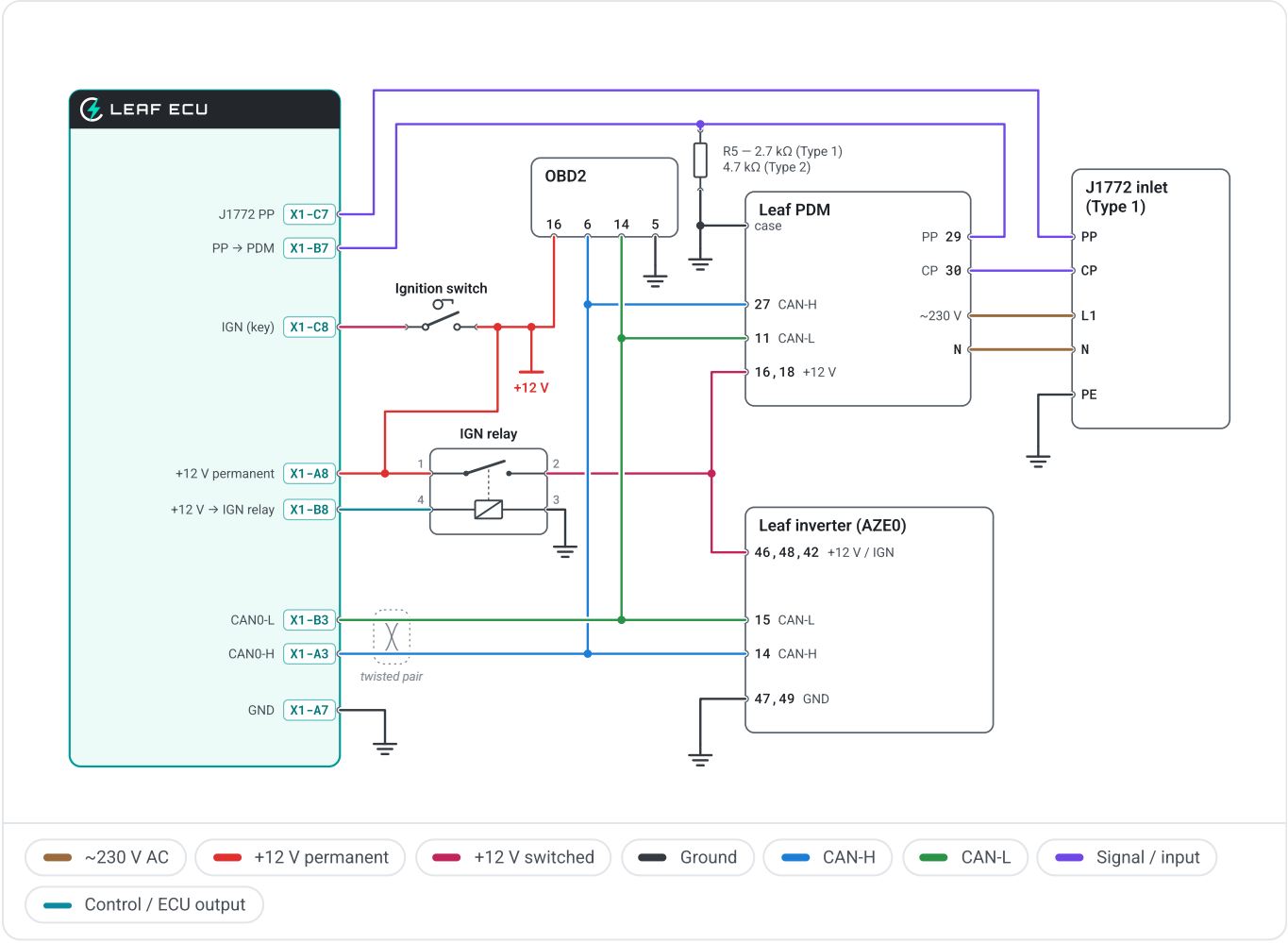


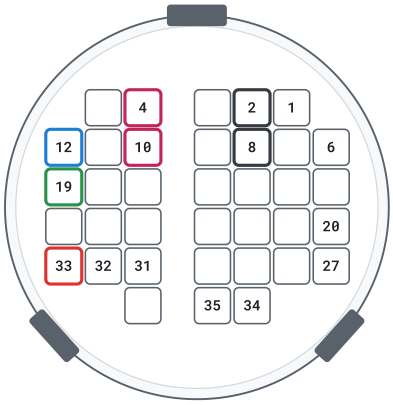
Fig. 3 V4 board 12 V supply: ignition switch, IGN relay, charge inlet and R5 resistor

The PP line between the PDM and the charge inlet must also be broken: the inlet PP goes to the controller (X1-C7), and the controller drives its own PP signal to the PDM (X1-B7). When a charging gun is plugged into the inlet, the controller wakes up and starts charging.

3 Inverter

To connect the inverter and the PDM it is best to reuse the factory Leaf harness: strip everything unnecessary, keeping the connectors for the inverter, the PDM and the motor resolver.

3.1 Leaf Inverter Gen 1 (ZE0)

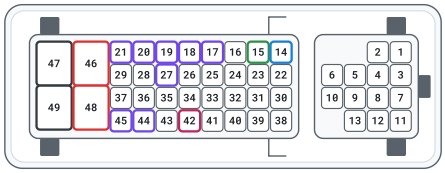


PIN	WIRE	FUNCTION
2		12 V ground
8		12 V ground
10		+12 V key switch (ignition)
4		+12 V key switch (ignition)
12		CAN-H (Powertrain, CAN0)
19		CAN-L (Powertrain, CAN0)
33		Always-on +12 V

Fig. 4 Leaf Gen 1 (ZE0) inverter — traction-motor-inverter connector

3.2 Leaf Inverter Gen 2 (AZE0)

Pins 46 and 48 are the permanent +12 V supply input, pin 42 is +12 V from the ignition switch, pins 47 and 49 are ground.

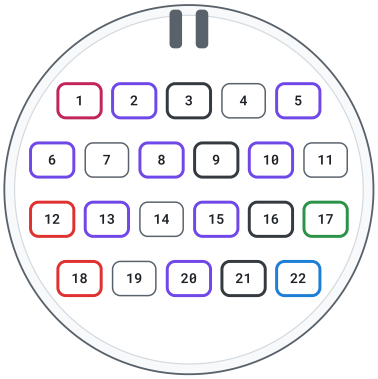


H.S. — harness (wire) side view

PIN	WIRE	FUNCTION
14		EV SYSTEM CAN-H — CAN-H (Powertrain, CAN0) · wire L (blue)
15		EV SYSTEM CAN-L — CAN-L (Powertrain, CAN0) · wire G (green)
17		REZ_S2 — resolver, winding S2 · wire P (pink)
18		REZ_S4 — resolver, winding S4 · wire L (blue)
19		REZ_R1 — resolver, excitation R1 · wire R (red)
20		REZ_S1 — resolver, winding S1 · wire B (black)
21		REZ_S3 — resolver, winding S3 · wire W (white)
27		REZ_R2 — resolver, excitation R2 · wire G (green)
42		IGN_SW — +12 V ignition · wire LG (light green)
44		TMGND — traction-motor temperature sensor ground · wire O (orange)
45		TM — traction-motor temperature sensor · wire Y (yellow)
46		VB1 — +12 V permanent, supply 1 · wire G (green)
47		VBGND1 — supply ground 1 · wire B (black)
48		VB2 — +12 V permanent, supply 2 · wire G (green)
49		VBGND2 — supply ground 2 · wire B (black)

Fig. 5 Leaf Gen 2 (AZE0) inverter — connector F13 (black)

3.3 Leaf Inverter Gen 3 (ZE1)



View looking into the panel-mount connector on the inverter

PIN	WIRE	FUNCTION
1		+12 V from the ignition switch (IGNSW) · red wire
2		Resolver S1 · light green wire
3		Ground · black wire
5		Resolver R1 — excitation · purple wire
6		Motor temperature sensor (MTEMP) · grey wire
8		Resolver S3 · light blue wire
9		Ground · black wire
10		Resolver R2 — excitation · brown wire
12		+12 V · red wire
13		Motor temperature sensor (MTEMP) · grey wire
15		Resolver S4 · blue wire
16		Ground · black wire
17		CAN-L (Powertrain, CAN0) · orange wire
18		+12 V · red wire
20		Resolver S2 · yellow wire
21		Ground · black wire
22		CAN-H (Powertrain, CAN0) · green wire

Fig. 6 Leaf Gen 3 (ZE1, 2018) inverter — control connector

Gen 2 ↔ Gen 3 pin map with the factory harness wire colours:

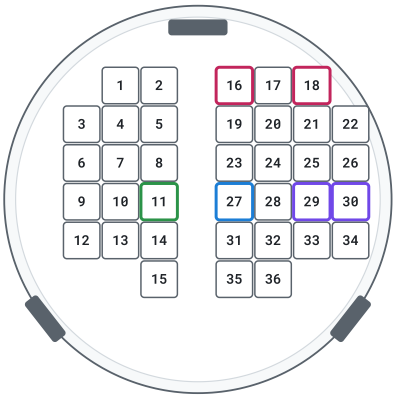
SIGNAL	GEN 2 PIN	GEN 2 COLOUR	GEN 3 PIN	GEN 3 COLOUR
CAN-H	14	L — Blue	5	L — Blue
CAN-L	15	G — Green	11	G — Green
Resolver S2	17	P — Pink	9	P — Pink
Resolver S4	18	L — Blue	3	L — Blue
Resolver R1	27	G — Green	16	G — Green
Resolver R2	19	R — Red	22	R — Red
Resolver S1	21	W — White	14	W — White
Resolver S3	20	B — Black	19	B — Black
12 V ignition	42	LG — Light green	18	LG — Light green
Traction motor temp	44	O — Orange	20	V — Violet
Traction motor temp	45	Y — Yellow	7	Y — Yellow

SIGNAL	GEN 2 PIN	GEN 2 COLOUR	GEN 3 PIN	GEN 3 COLOUR
12 V	**46**	G — Green	**1**	G — Green
Ground	**47**	B — Black	**4**	B — Black
12 V	**48**	G — Green	**6**	G — Green
Ground	**49**	B — Black	**10**	B — Black

WARNING

CAN-H and CAN-L wires must be twisted into a twisted pair.

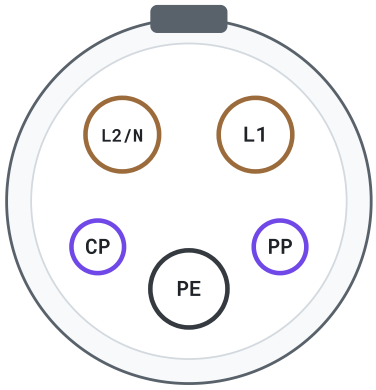
4 PDM Module



Harness side view (H.S.)

PIN	WIRE	FUNCTION
11		CAN-L (Powertrain, CAN0)
16		+12 V PDM supply (V4 board: from the IGN relay)
18		+12 V PDM supply (V4 board: from the IGN relay)
27		CAN-H (Powertrain, CAN0)
29		Charge-inlet PP (V4 board: from ECU X1-B7)
30		Charge-inlet CP (Control Pilot)

Fig. 7 PDM module interface connector (F23)



Front view of the vehicle inlet

PIN	WIRE	FUNCTION	DIAGRAMS
CP		Control Pilot → PDM (pin 30)	2 3
L1		Phase L1 (~230 V) → PDM	2 3
N		Neutral N (L2) → PDM	2 3
PE		Protective earth → chassis	2 3
PP		Proximity Pilot → ECU X1-C7	2 3

Fig. 8 J1772 (Type 1) charge inlet: vehicle socket

PDM pins 30 and 29 (charge inlet) connect to the two-pin connector on the inlet itself: white wire to white, green to brown (when using the Leaf harness). With the V4 board wire the PP line as shown in Fig. 3.

The PDM case must be connected to ground — the case is the negative output of the DC-DC converter. The stud on the side of the module is the +12 V output of the DC-DC converter.

> **TYPE 2**
> If a Type 2 inlet is used instead of Type 1, resistor R5 must be changed (see Fig. 3).

5 Controls

5.1 Selector Buttons

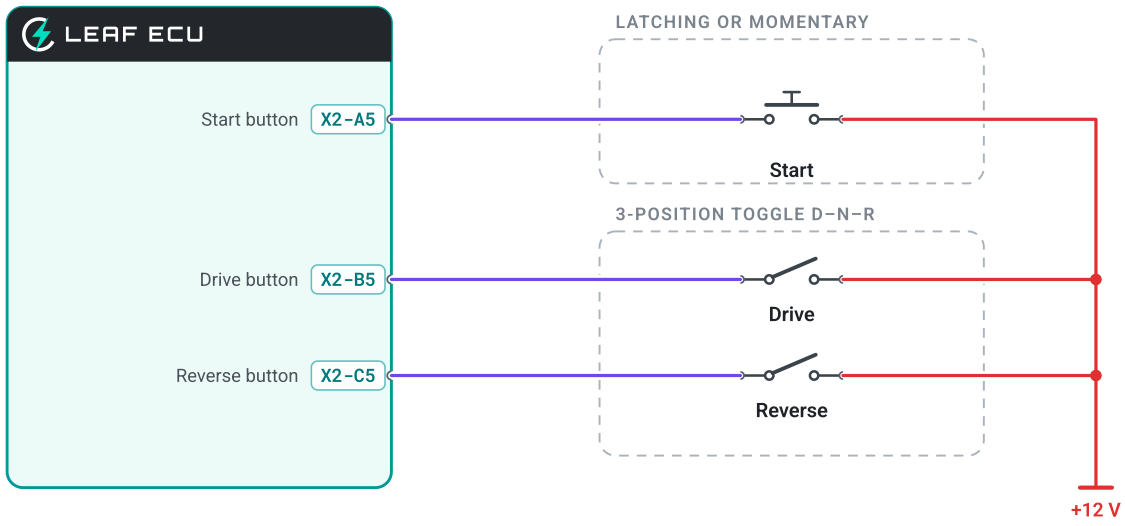


Fig. 9 Start, Drive and Reverse control buttons

Driving direction (the selector) can be chosen with buttons or with the original Nissan Leaf selector. The selector type must be set in the configurator.

- **Drive and Reverse buttons** — latching.
- **Start button** — latching or momentary; its mode is set in the configurator.

How it works:

1. Pressing **Start** closes the contactors; the inverter is ready, the selector is in neutral.
2. To drive off, switch on **Drive** or **Reverse**.
3. If Drive and Reverse are active at the same time, the controller goes to neutral.

RECOMMENDATION

Use a 3-position toggle switch for Drive and Reverse: centre position = neutral, end positions = Drive and Reverse.

5.2 Brake Vacuum Booster

The controller can run the vacuum pump of the brake booster. Connect the vacuum sensor to the controller and the pump through a relay.

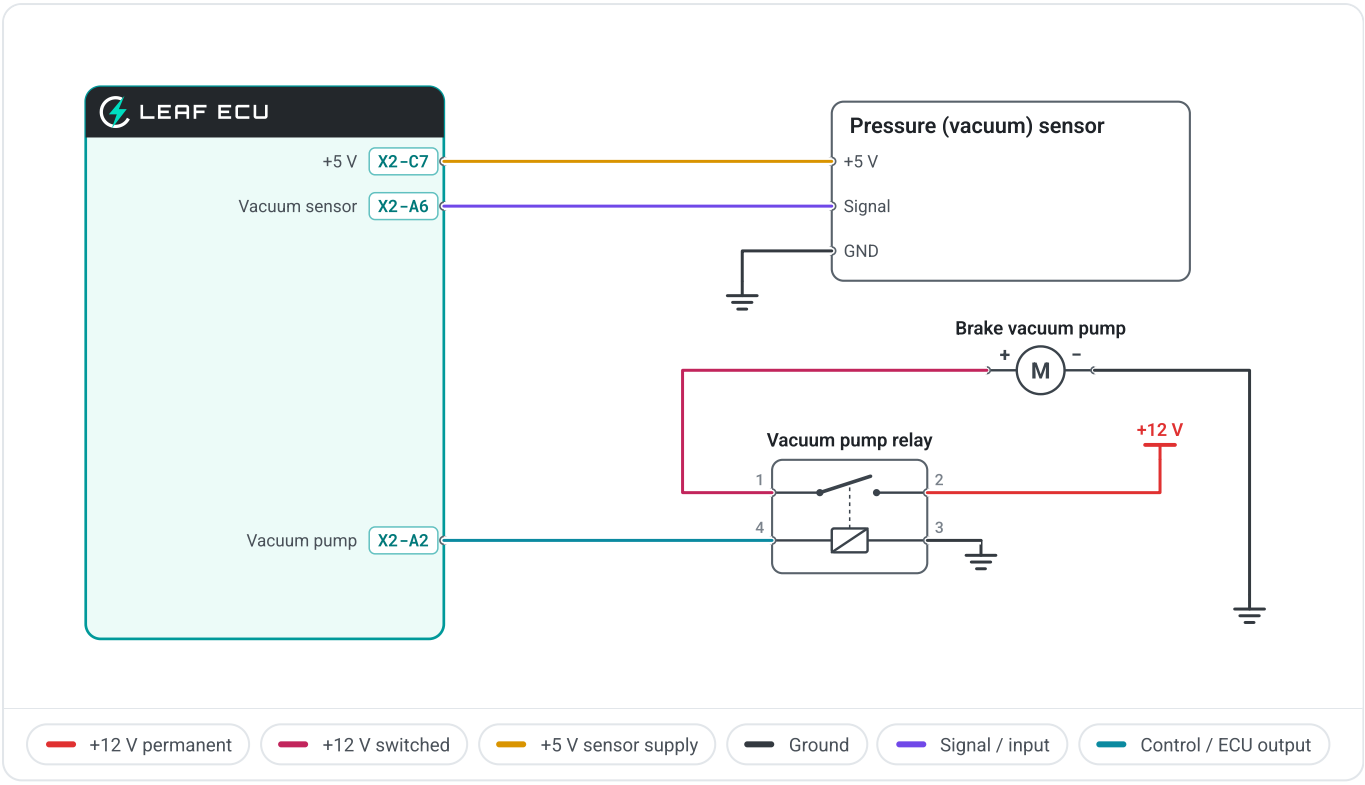


Fig. 10 Brake vacuum booster: vacuum sensor and vacuum pump relay

The sensor voltage thresholds that switch the pump on and off are set in the configurator.

6 Power Relays (Contactors)

Choose power relays with ratings close to those used in the Nissan Leaf.

TESLA CONTACTORS ARE NOT SUITABLE

They have a high coil holding current and need an economiser. Similar contactors with a built-in current-reduction circuit exist — those can be used.

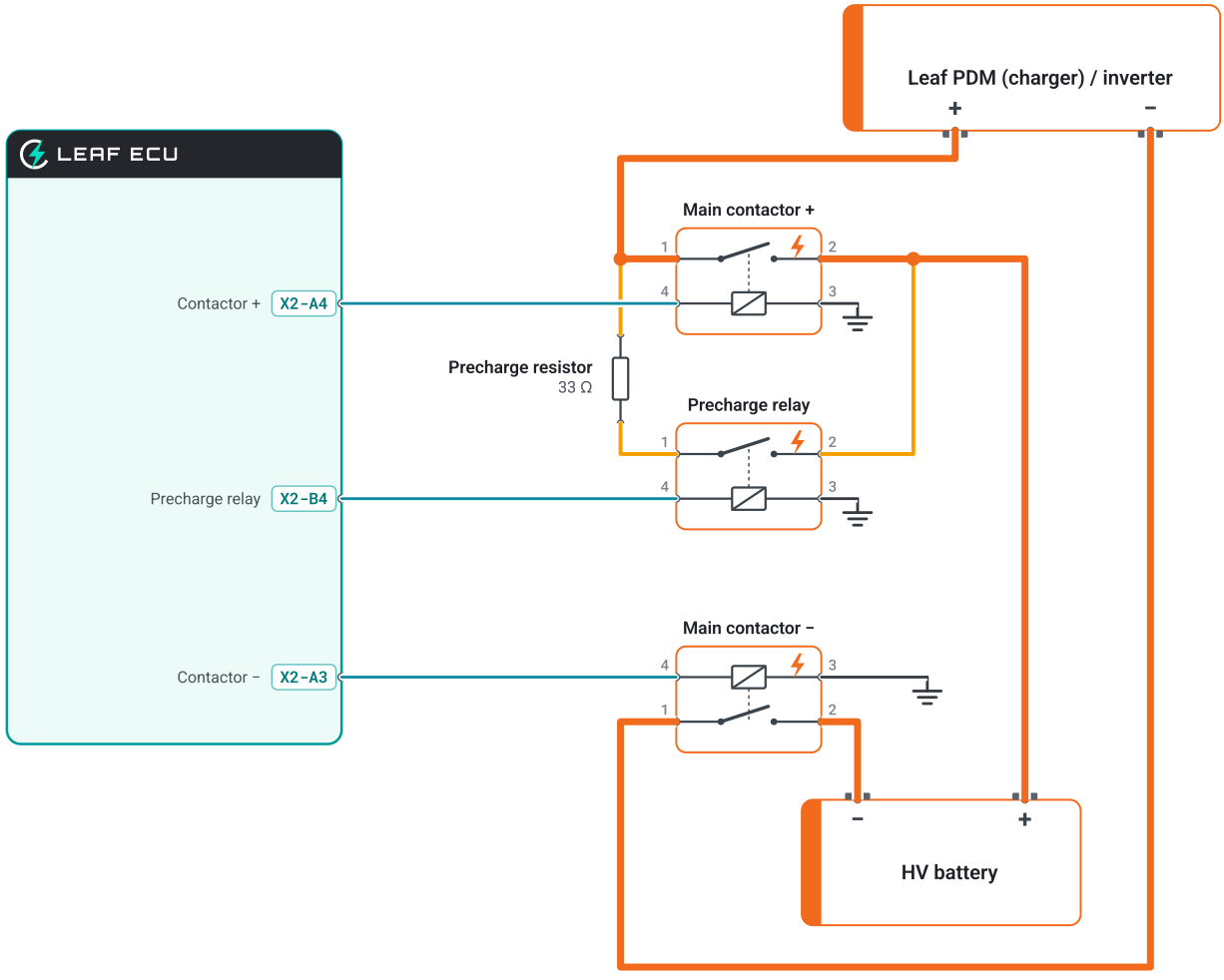


Fig. 11 Power contactors and precharge: main contactors + and −, precharge relay and resistor

Wire cross-sections in the diagram:

WIRE	CROSS-SECTION	PURPOSE
Power	35 mm²	Main traction battery cables
Precharge	2.5 mm²	Precharge path through the 33 Ω resistor
Control	0.75 mm²	Relay coils

Observe polarity when connecting the traction battery to the PDM:

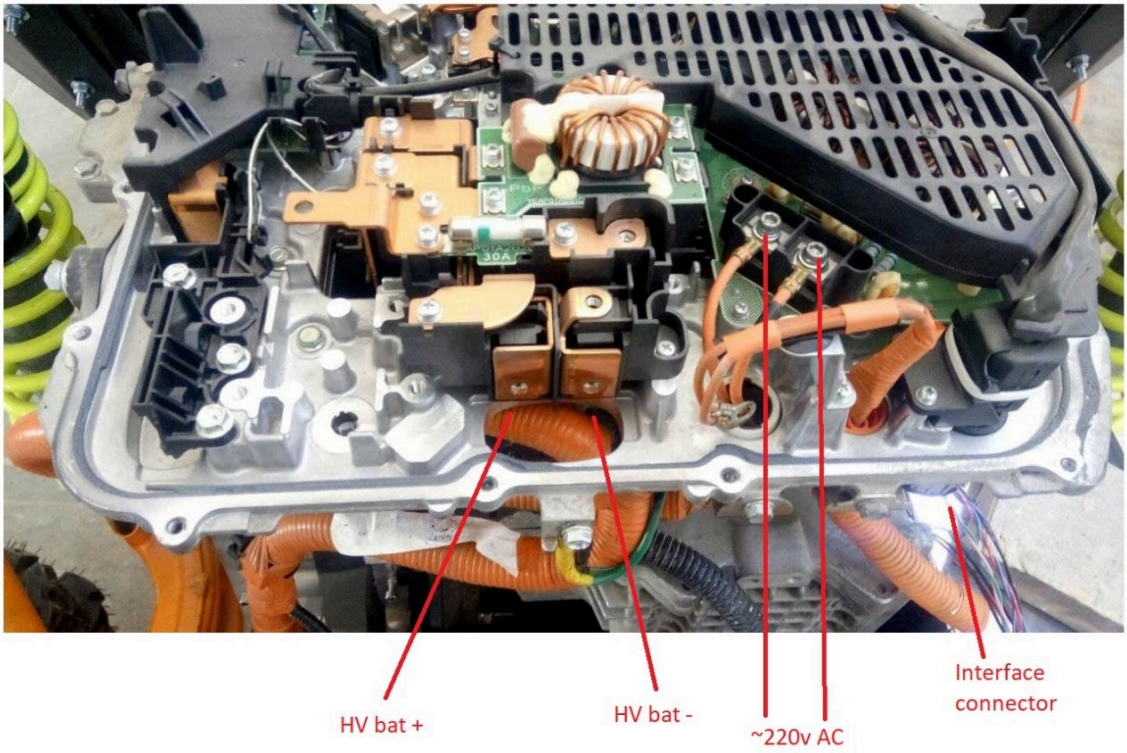


Fig. 12 Traction battery polarity at the PDM: HV bat + / HV bat −, ~230 V input and interface connector

7 Accelerator Pedal

The pedal signal wire connects to pin X1-B2.

PIN	FUNCTION
X1-B2	Pedal signal — primary sensor
X1-A2	Pedal signal — second sensor (not used by the firmware yet)
X1-C1	+5 V pedal supply
X1-C2	Pedal ground

The signal voltage must swing within 0–5 V — check this before connecting to the controller.

TIP

If your pedal has a second sensor, wire it to X1-A2 right away.

Both resistive and Hall-effect pedals can be used.

WARNING

When connecting a Hall-effect pedal, double-check the sensor supply polarity and which wire is the signal output — otherwise you can damage the sensor or the controller's 5 V regulator.

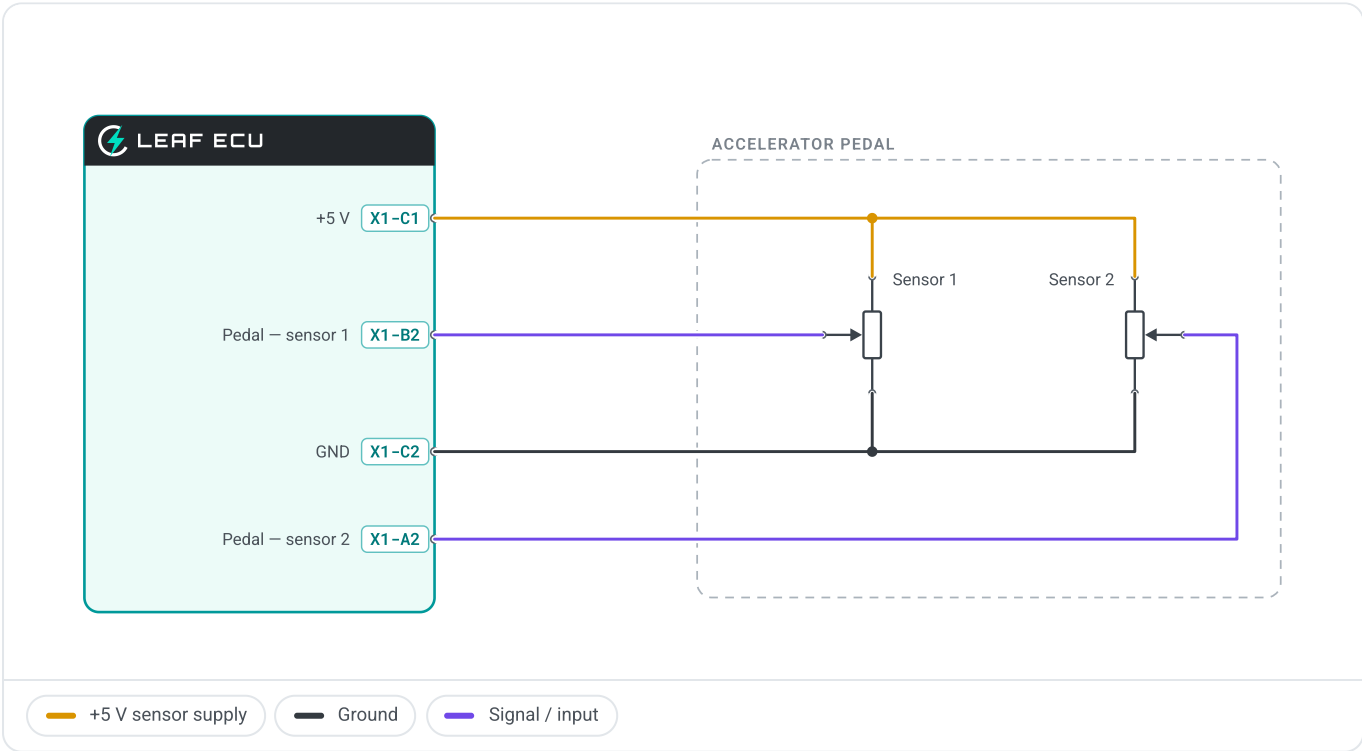


Fig. 13 Accelerator pedal wiring (two potentiometer tracks)



Fig. 14 Nissan Leaf accelerator pedal



PIN	WIRE	FUNCTION
1		Sensor 2 — signal
2		Sensor 2 — supply "+"
3		Sensor 1 — supply "+"
4		Sensor 1 — signal
5		Sensor 1 — ground "−"
6		Sensor 2 — ground "−"

Fig. 15 Nissan Leaf accelerator pedal connector (the same connector is used in other Nissan and Renault models)

The minimum and maximum pedal voltages must be set in the configurator (see 16.2).

8 BMS

The controller supports the Chevrolet Volt Gen 1 BMS, Orion 2 and the Nissan Leaf BMS.

8.1 Chevrolet Volt Gen 1

The BMS consists of 4 satellite modules marked 1, 2/3 (two modules) and 4. With the satellites the battery can be split into up to four separate packs.

> **NOTE**
> Satellites 2/3 have a jumper that sets the satellite's position in the battery. Without it the satellite will not work.

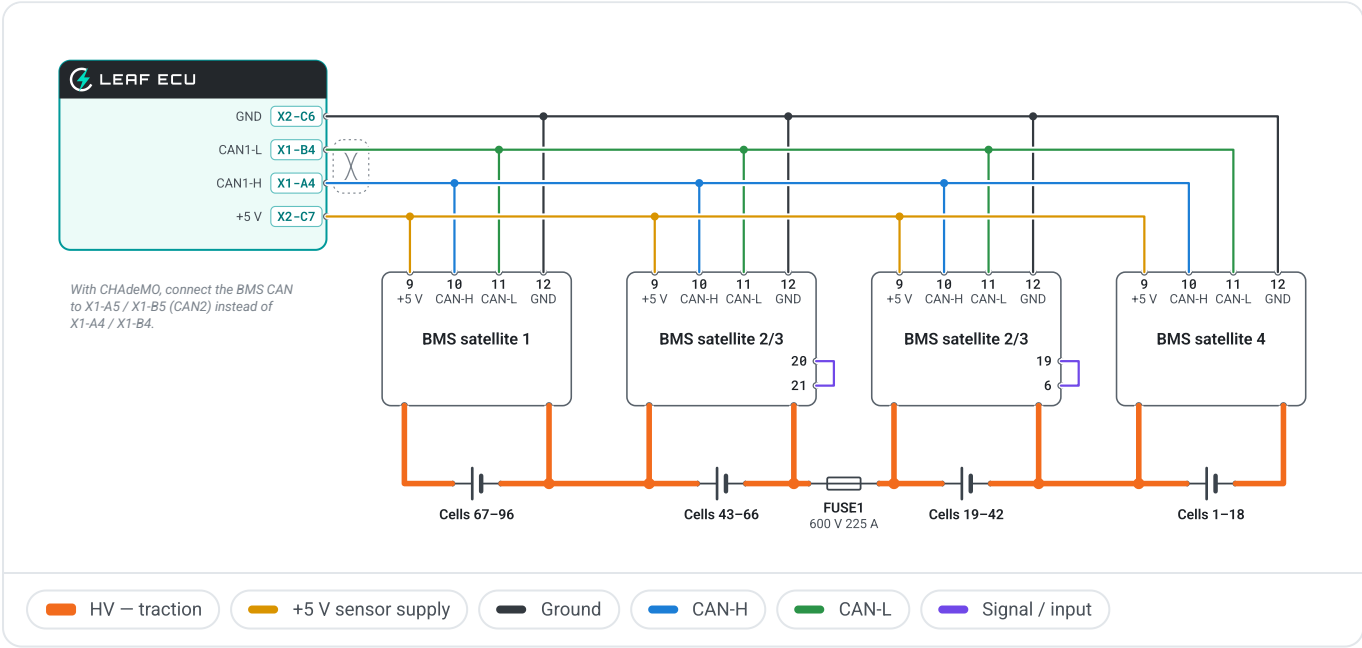
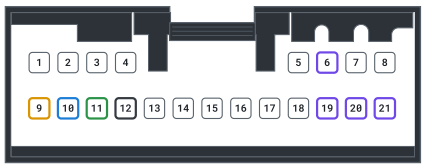


Fig. 16 Chevrolet Volt Gen 1 BMS: connecting the four satellites to the ECU and the battery cell split

The **black connector** on a satellite is the interface connector. We recommend wiring CAN and power as shown in the diagram rather than through the Volt factory harness, where wires pass through the satellites in a daisy chain, which makes diagnostics harder.



View from the BMS (satellite) side

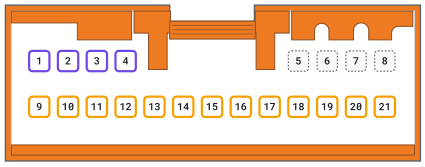
PIN	WIRE	FUNCTION	DIAGRAMS
6		Jumper 19–6 — sets the position of satellite 2/3 (cell group 19–42)	16
9		+5 V — satellite supply (ECU X2-C7)	16
10		CAN-H (ECU X1-A4; with CHAdeMO — X1-A5)	16
11		CAN-L (ECU X1-B4; with CHAdeMO — X1-B5)	16
12		Ground (ECU X2-C6)	16
19		Jumper 19–6 — sets the position of satellite 2/3 (cell group 19–42)	16
20		Jumper 20–21 — sets the position of satellite 2/3 (cell group 43–66)	16
21		Jumper 20–21 — sets the position of satellite 2/3 (cell group 43–66)	16

Fig. 17 Volt BMS satellite — interface connector (black)

The **orange connectors** on the satellites are for the cell balancing wires and temperature sensors. This BMS requires **NTC 10 kΩ** temperature sensors.

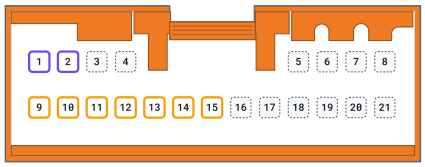
There are two connector types: 12-cell (satellites 2/3 have only this type) and 6-cell (found on satellites 1 and 4).

Volt BMS satellite — cell / temperature connector, 12 cells (orange)



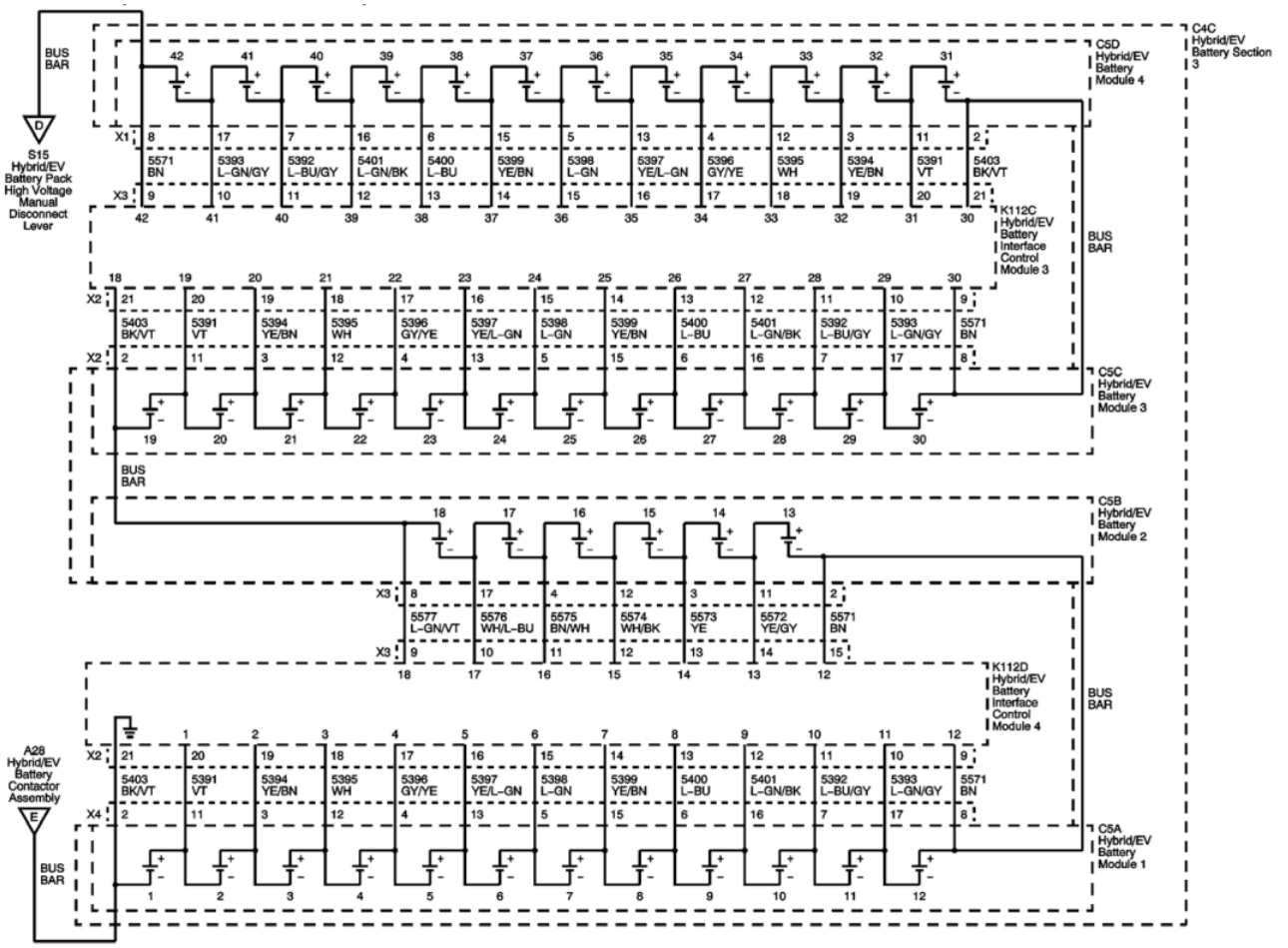
PIN	WIRE	FUNCTION
1		Battery module temperature signal (NTC 10 kΩ)
2		Low reference — temperature sensor return
3		Battery module temperature signal (NTC 10 kΩ)
4		Low reference — temperature sensor return
5		Not used
6		Not used
7		Not used
8		Not used
9		Battery voltage signal (12)
10		Battery voltage signal (11)
11		Battery voltage signal (10)
12		Battery voltage signal (9)
13		Battery voltage signal (8)
14		Battery voltage signal (7)
15		Battery voltage signal (6)
16		Battery voltage signal (5)
17		Battery voltage signal (4)
18		Battery voltage signal (3)
19		Battery voltage signal (2)
20		Battery voltage signal (1)
21		Low reference — bottom sense point of the cell voltages

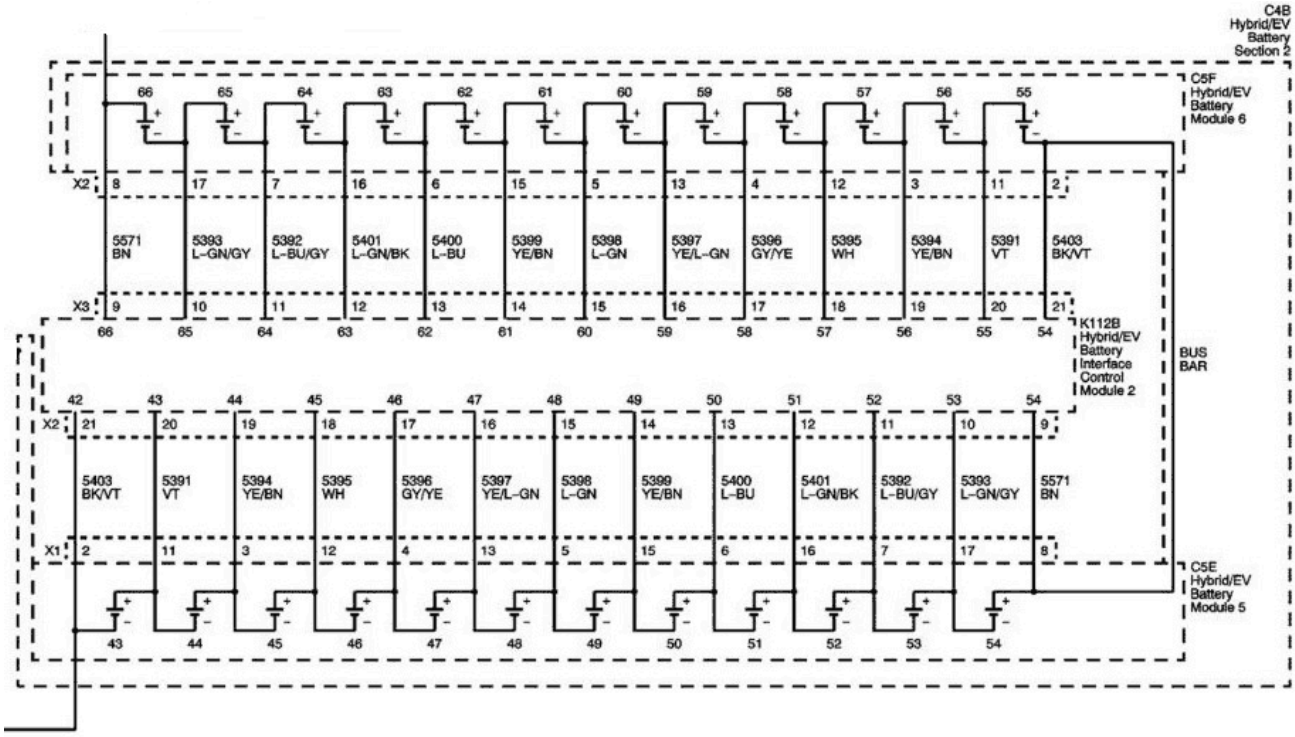
Volt BMS satellite — cell / temperature connector, 6 cells (orange)



PIN	WIRE	FUNCTION
1		Battery module temperature signal (NTC 10 kΩ)
2		Low reference — temperature sensor return
3		Not used
4		Not used
5		Not used
6		Not used
7		Not used
8		Not used
9		Battery voltage signal (18)
10		Battery voltage signal (17)
11		Battery voltage signal (16)
12		Battery voltage signal (15)
13		Battery voltage signal (14)
14		Battery voltage signal (13)
15		Battery voltage signal (12)
16		Not used
17		Not used
18		Not used
19		Not used
20		Not used
21		Not used

Fig. 18 Volt satellite cell connectors: 12-cell and 6-cell variants


BUS BAR
S15 Hybrid/EV Battery Pack High Voltage Manual Disconnect Lever
C4C Hybrid/EV Battery Section 3
C5D Hybrid/EV Battery Module 4
C5C Hybrid/EV Battery Module 3
C5B Hybrid/EV Battery Module 2
C5A Hybrid/EV Battery Module 1
K112C Hybrid/EV Battery Interface Control Module 3
K112D Hybrid/EV Battery Interface Control Module 4
A28 Hybrid/EV Battery Contactor Assembly
X1
X2
X3
X4
5571 BN
5393 L-GN/GY
5392 L-BU/GY
5401 L-GN/BK
5400 L-BU
5399 YE/BN
5398 L-GN
5397 YE/L-GN
5396 GY/YE
5395 WH
5394 YE/BN
5391 VT
5403 BK/VT
5577 L-GN/VT
5576 WH/L-BU
5575 BN/WH
5574 WH/BK
5573 YE
5572 YE/GY


C4B Hybrid/EV Battery Section 2
C5F Hybrid/EV Battery Module 6
C5E Hybrid/EV Battery Module 5
K112B Hybrid/EV Battery Interface Control Module 2
BUS BAR
X1
X2
X3
5571 BN
5393 L-GN/GY
5392 L-BU/GY
5401 L-GN/BK
5400 L-BU
5399 YE/BN
5398 L-GN
5397 YE/L-GN
5396 GY/YE
5395 WH
5394 YE/BN
5391 VT
5403 BK/VT

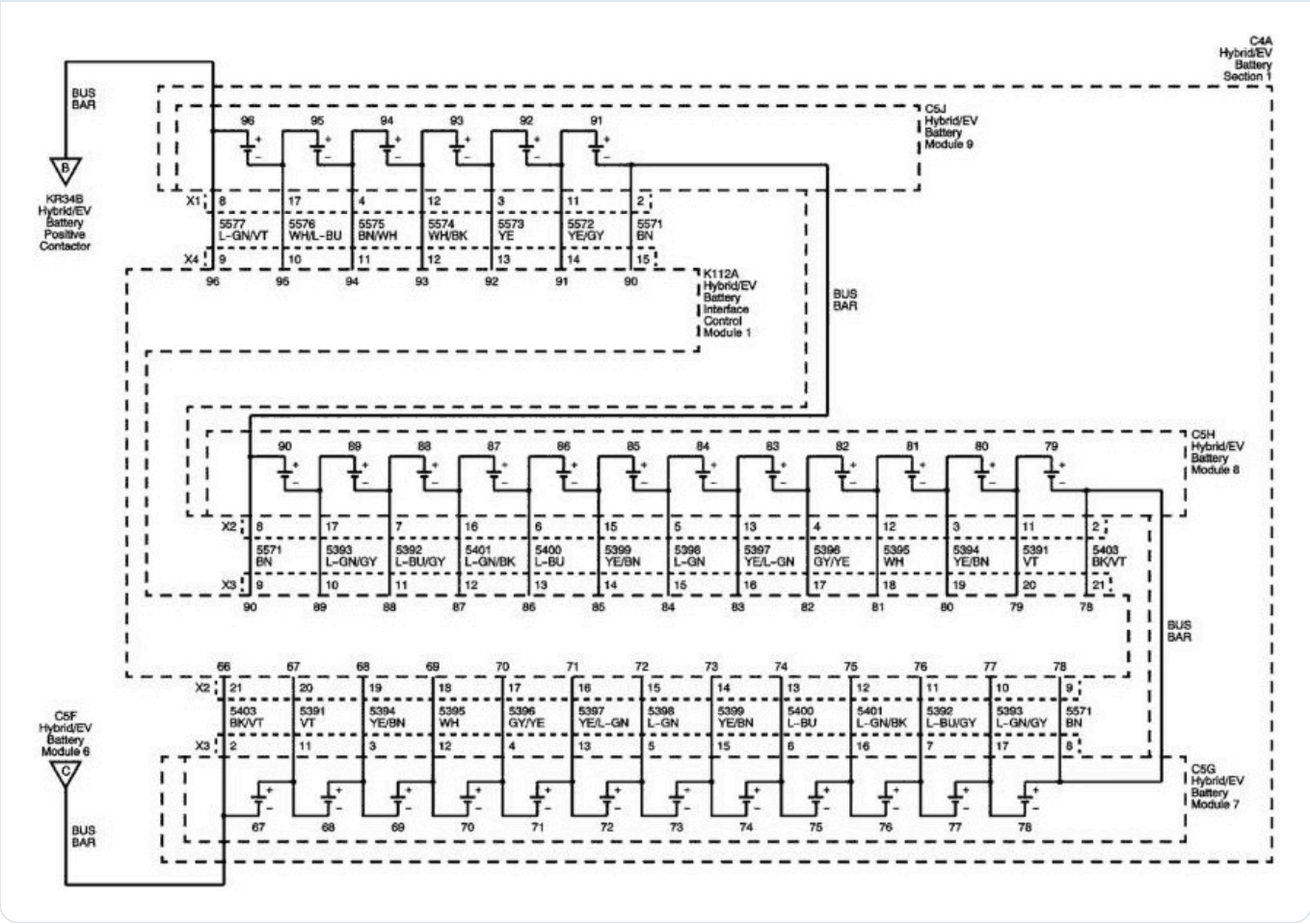


Fig. 19 Satellite placement sequence in the battery — original GM drawing

> **NOTE**
>
> The satellite sequence can be changed, but the cell map will then show positions incorrectly.

8.2 Nissan Leaf (ZE0 / AZE0)

The Nissan Leaf AZE0 and ZE0 BMS can be used (ZE1 is being tested). The BMS connects over CAN:

CONFIGURATION	CAN-H	CAN-L
Without CHAdeMO	X1-A4	X1-B4
With CHAdeMO	X1-A5	X1-B5

The original temperature sensors and the current sensor must also be connected to the BMS. The interlocks can be left out.

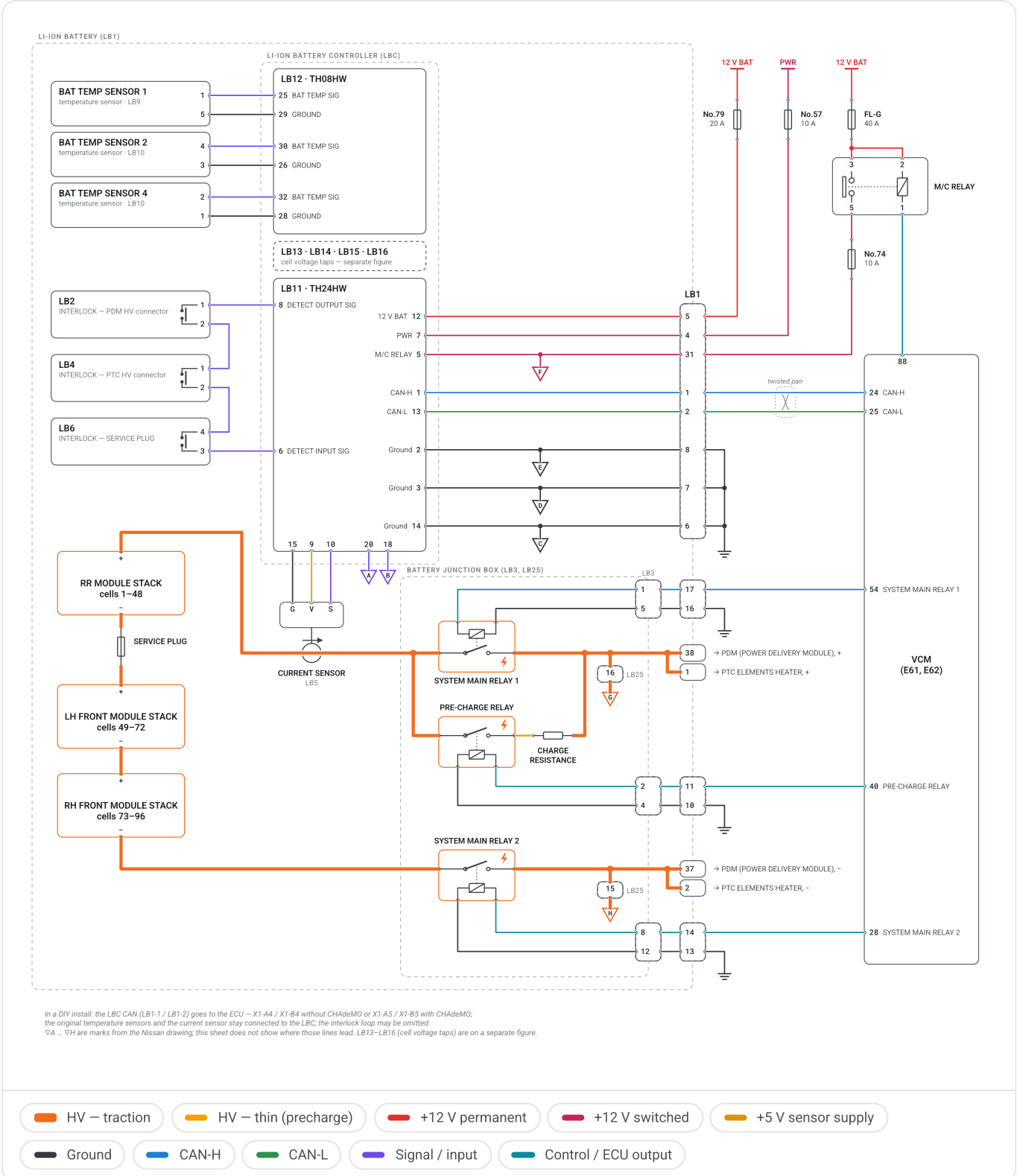


Fig. 20 Leaf AZE0 traction battery: LBC external circuits — 12 V supply, CAN, ground, temperature and current sensors, interlocks, battery junction box (SMRs, pre-charge) and LB1 pack-connector pins

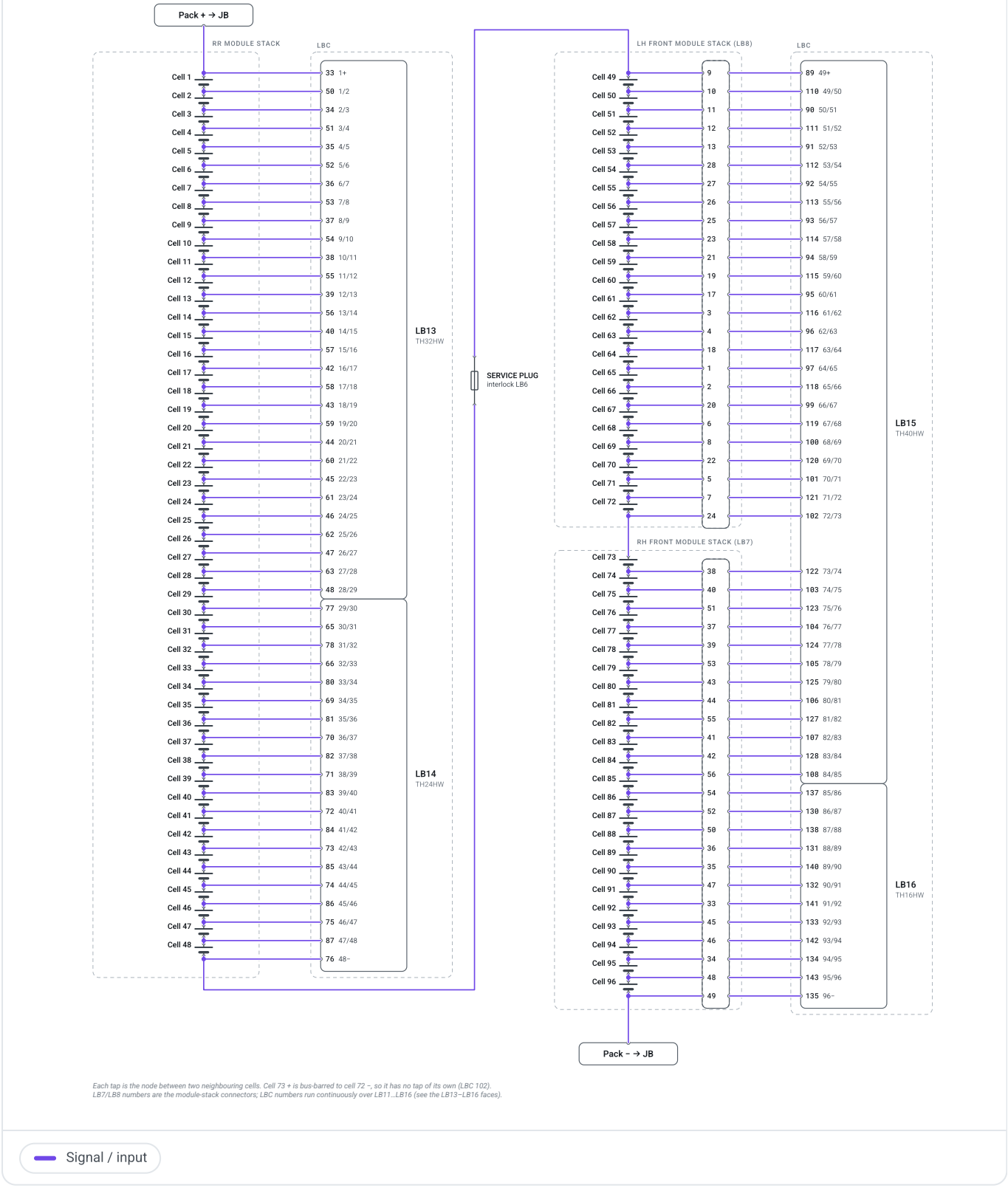


Fig. 21 Leaf AZE0 traction battery: voltage taps of the 96 cells → LBC (LB13–LB16) via stack connectors LB8 and LB7

LBC — connector LB11 (TH24HW): supply, CAN, ground, current sensor, interlock



PIN	WIRE	FUNCTION	DIAGRAMS
1		CAN-H → LB1-1 → VCM 24 (DIY: ECU X1-A4 / X1-A5)	20
2		Ground → LB1-8 (branch ▽E)	20
3		Ground → LB1-7 (branch ▽D)	20
5		+12 V from the M/C RELAY via fuse No.74 10 A → LB1-31 (branch ▽F)	20
6		DETECT INPUT SIG — interlock loop input (from LB6-3)	20
7		+12 V "PWR" via fuse No.57 10 A → LB1-4	20
8		DETECT OUTPUT SIG — interlock loop output (to LB2-1)	20
9		VOLTAGE — current sensor supply (LB5 "V")	20
10		CURRENT SIG — current sensor signal (LB5 "S")	20
12		+12 V "12 V BAT" via fuse No.79 20 A → LB1-5	20
13		CAN-L → LB1-2 → VCM 25 (DIY: ECU X1-B4 / X1-B5)	20
14		Ground → LB1-6 (branch ▽C)	20
15		GROUND — current sensor ground (LB5 "G")	20
18		Line to ▽B (destination not shown on the drawing)	20
20		Line to ▽A (destination not shown on the drawing)	20

LBC — connector LB12 (TH08HW): battery temperature sensors



PIN	WIRE	FUNCTION	DIAGRAMS
25		BAT TEMP SIG — sensor 1 (LB9-1)	20
26		GROUND — sensor 2 (LB10-3)	20
28		GROUND — sensor 4 (LB10-1)	20
29		GROUND — sensor 1 (LB9-5)	20
30		BAT TEMP SIG — sensor 2 (LB10-4)	20
32		BAT TEMP SIG — sensor 4 (LB10-2)	20

LBC — connector LB13 (TH32HW): cell voltage taps (RR stack, cells 1–29)



PIN	WIRE	FUNCTION	DIAGRAMS
33		Cell 1 + (pack +) — voltage tap (RR module stack)	21
34		Cells 2/3 — voltage tap (RR module stack)	21
35		Cells 4/5 — voltage tap (RR module stack)	21
36		Cells 6/7 — voltage tap (RR module stack)	21
37		Cells 8/9 — voltage tap (RR module stack)	21
38		Cells 10/11 — voltage tap (RR module stack)	21
39		Cells 12/13 — voltage tap (RR module stack)	21
40		Cells 14/15 — voltage tap (RR module stack)	21
42		Cells 16/17 — voltage tap (RR module stack)	21
43		Cells 18/19 — voltage tap (RR module stack)	21
44		Cells 20/21 — voltage tap (RR module stack)	21
45		Cells 22/23 — voltage tap (RR module stack)	21
46		Cells 24/25 — voltage tap (RR module stack)	21
47		Cells 26/27 — voltage tap (RR module stack)	21
48		Cells 28/29 — voltage tap (RR module stack)	21
50		Cells 1/2 — voltage tap (RR module stack)	21
51		Cells 3/4 — voltage tap (RR module stack)	21
52		Cells 5/6 — voltage tap (RR module stack)	21
53		Cells 7/8 — voltage tap (RR module stack)	21
54		Cells 9/10 — voltage tap (RR module stack)	21
55		Cells 11/12 — voltage tap (RR module stack)	21
56		Cells 13/14 — voltage tap (RR module stack)	21
57		Cells 15/16 — voltage tap (RR module stack)	21
58		Cells 17/18 — voltage tap (RR module stack)	21
59		Cells 19/20 — voltage tap (RR module stack)	21
60		Cells 21/22 — voltage tap (RR module stack)	21
61		Cells 23/24 — voltage tap (RR module stack)	21
62		Cells 25/26 — voltage tap (RR module stack)	21
63		Cells 27/28 — voltage tap (RR module stack)	21

LBC — connector LB14 (TH24HW): cell voltage taps (RR stack, cells 29–48)



PIN	WIRE	FUNCTION	DIAGRAMS
65		Cells 30/31 — voltage tap (RR module stack)	21
66		Cells 32/33 — voltage tap (RR module stack)	21
69		Cells 34/35 — voltage tap (RR module stack)	21
70		Cells 36/37 — voltage tap (RR module stack)	21
71		Cells 38/39 — voltage tap (RR module stack)	21
72		Cells 40/41 — voltage tap (RR module stack)	21
73		Cells 42/43 — voltage tap (RR module stack)	21
74		Cells 44/45 — voltage tap (RR module stack)	21
75		Cells 46/47 — voltage tap (RR module stack)	21
76		Cell 48 − (to the service plug) — voltage tap (RR module stack)	21
77		Cells 29/30 — voltage tap (RR module stack)	21
78		Cells 31/32 — voltage tap (RR module stack)	21
80		Cells 33/34 — voltage tap (RR module stack)	21
81		Cells 35/36 — voltage tap (RR module stack)	21
82		Cells 37/38 — voltage tap (RR module stack)	21
83		Cells 39/40 — voltage tap (RR module stack)	21
84		Cells 41/42 — voltage tap (RR module stack)	21
85		Cells 43/44 — voltage tap (RR module stack)	21
86		Cells 45/46 — voltage tap (RR module stack)	21
87		Cells 47/48 — voltage tap (RR module stack)	21

LBC — connector LB15 (TH40HW): cell voltage taps (LH stack cells 49–72 and RH stack cells 73–85)



PIN	WIRE	FUNCTION	DIAGRAMS
89		Cell 49 + (after the service plug) — voltage tap (LH front stack, via LB8-9)	21
90		Cells 50/51 — voltage tap (LH front stack, via LB8-11)	21
91		Cells 52/53 — voltage tap (LH front stack, via LB8-13)	21
92		Cells 54/55 — voltage tap (LH front stack, via LB8-27)	21
93		Cells 56/57 — voltage tap (LH front stack, via LB8-25)	21
94		Cells 58/59 — voltage tap (LH front stack, via LB8-21)	21
95		Cells 60/61 — voltage tap (LH front stack, via LB8-17)	21
96		Cells 62/63 — voltage tap (LH front stack, via LB8-4)	21
97		Cells 64/65 — voltage tap (LH front stack, via LB8-1)	21
99		Cells 66/67 — voltage tap (LH front stack, via LB8-20)	21
100		Cells 68/69 — voltage tap (LH front stack, via LB8-8)	21
101		Cells 70/71 — voltage tap (LH front stack, via LB8-5)	21
102		Cells 72/73 — voltage tap (LH front stack, via LB8-24)	21
103		Cells 74/75 — voltage tap (RH front stack, via LB7-40)	21
104		Cells 76/77 — voltage tap (RH front stack, via LB7-37)	21
105		Cells 78/79 — voltage tap (RH front stack, via LB7-53)	21
106		Cells 80/81 — voltage tap (RH front stack, via LB7-44)	21
107		Cells 82/83 — voltage tap (RH front stack, via LB7-41)	21
108		Cells 84/85 — voltage tap (RH front stack, via LB7-56)	21
110		Cells 49/50 — voltage tap (LH front stack, via LB8-10)	21
111		Cells 51/52 — voltage tap (LH front stack, via LB8-12)	21
112		Cells 53/54 — voltage tap (LH front stack, via LB8-28)	21
113		Cells 55/56 — voltage tap (LH front stack, via LB8-26)	21
114		Cells 57/58 — voltage tap (LH front stack, via LB8-23)	21
115		Cells 59/60 — voltage tap (LH front stack, via LB8-19)	21
116		Cells 61/62 — voltage tap (LH front stack, via LB8-3)	21
117		Cells 63/64 — voltage tap (LH front stack, via LB8-18)	21
118		Cells 65/66 — voltage tap (LH front stack, via LB8-2)	21
119		Cells 67/68 — voltage tap (LH front stack, via LB8-6)	21
120		Cells 69/70 — voltage tap (LH front stack, via LB8-22)	21
121		Cells 71/72 — voltage tap (LH front stack, via LB8-7)	21
122		Cells 73/74 — voltage tap (RH front stack, via LB7-38)	21
123		Cells 75/76 — voltage tap (RH front stack, via LB7-51)	21
124		Cells 77/78 — voltage tap (RH front stack, via LB7-39)	21
125		Cells 79/80 — voltage tap (RH front stack, via LB7-43)	21
127		Cells 81/82 — voltage tap (RH front stack, via LB7-55)	21

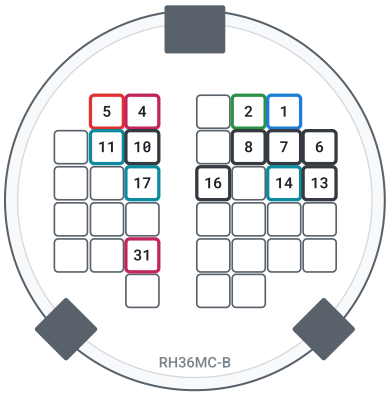
PIN	WIRE	FUNCTION	DIAGRAMS
128		Cells 83/84 — voltage tap (RH front stack, via LB7-42)	21

LBC — connector LB16 (TH16HW): cell voltage taps (RH stack, cells 85–96)



PIN	WIRE	FUNCTION	DIAGRAMS
130		Cells 86/87 — voltage tap (RH front stack, via LB7-52)	21
131		Cells 88/89 — voltage tap (RH front stack, via LB7-36)	21
132		Cells 90/91 — voltage tap (RH front stack, via LB7-47)	21
133		Cells 92/93 — voltage tap (RH front stack, via LB7-45)	21
134		Cells 94/95 — voltage tap (RH front stack, via LB7-34)	21
135		Cell 96 − (pack −) — voltage tap (RH front stack, via LB7-49)	21
137		Cells 85/86 — voltage tap (RH front stack, via LB7-54)	21
138		Cells 87/88 — voltage tap (RH front stack, via LB7-50)	21
140		Cells 89/90 — voltage tap (RH front stack, via LB7-35)	21
141		Cells 91/92 — voltage tap (RH front stack, via LB7-33)	21
142		Cells 93/94 — voltage tap (RH front stack, via LB7-46)	21
143		Cells 95/96 — voltage tap (RH front stack, via LB7-48)	21

Fig. 22 Li-ion battery controller (LBC) connectors LB11–LB16



PIN	WIRE	FUNCTION	DIAGRAMS
1		CAN-H: LBC 1 ↔ VCM 24 (DIY: ECU X1-A4 without CHAdeMO / X1-A5 with CHAdeMO)	20
2		CAN-L: LBC 13 ↔ VCM 25 (DIY: ECU X1-B4 / X1-B5)	20
4		+12 V "PWR" (fuse No.57 10 A) → LBC 7	20
5		+12 V "12 V BAT" (fuse No.79 20 A) → LBC 12	20
31		+12 V from the M/C RELAY (fuse No.74 10 A) → LBC 5	20
6		Ground, LBC 14	20
7		Ground, LBC 3	20
8		Ground, LBC 2	20
17		SYSTEM MAIN RELAY 1 coil (LB3-1) ← VCM 54	20
16		SYSTEM MAIN RELAY 1 coil ground (LB3-5)	20
14		SYSTEM MAIN RELAY 2 coil (LB3-8) ← VCM 28	20
13		SYSTEM MAIN RELAY 2 coil ground (LB3-12)	20
11		PRE-CHARGE RELAY coil (LB3-2) ← VCM 40	20
10		PRE-CHARGE RELAY coil ground (LB3-4)	20

Fig. 23 Battery connector LB1

LB8 (TH24FW-NH) — LH front module stack connector (cells 49–72)



PIN	WIRE	FUNCTION	DIAGRAMS
1		Cells 64/65 → LBC 97	21
2		Cells 65/66 → LBC 118	21
3		Cells 61/62 → LBC 116	21
4		Cells 62/63 → LBC 96	21
5		Cells 70/71 → LBC 101	21
6		Cells 67/68 → LBC 119	21
7		Cells 71/72 → LBC 121	21
8		Cells 68/69 → LBC 100	21
9		Cell 49 + (after the service plug) → LBC 89	21
10		Cells 49/50 → LBC 110	21
11		Cells 50/51 → LBC 90	21
12		Cells 51/52 → LBC 111	21
13		Cells 52/53 → LBC 91	21
17		Cells 60/61 → LBC 95	21
18		Cells 63/64 → LBC 117	21
19		Cells 59/60 → LBC 115	21
20		Cells 66/67 → LBC 99	21
21		Cells 58/59 → LBC 94	21
22		Cells 69/70 → LBC 120	21
23		Cells 57/58 → LBC 114	21
24		Cells 72/73 → LBC 102	21
25		Cells 56/57 → LBC 93	21
26		Cells 55/56 → LBC 113	21
27		Cells 54/55 → LBC 92	21
28		Cells 53/54 → LBC 112	21

LB7 (TH32FW-NH) — RH front module stack connector (cells 73–96)



PIN	WIRE	FUNCTION	DIAGRAMS
33		Cells 91/92 → LBC 141	21
34		Cells 94/95 → LBC 134	21
35		Cells 89/90 → LBC 140	21
36		Cells 88/89 → LBC 131	21
37		Cells 76/77 → LBC 104	21
38		Cells 73/74 → LBC 122	21
39		Cells 77/78 → LBC 124	21
40		Cells 74/75 → LBC 103	21
41		Cells 82/83 → LBC 107	21
42		Cells 83/84 → LBC 128	21
43		Cells 79/80 → LBC 125	21
44		Cells 80/81 → LBC 106	21
45		Cells 92/93 → LBC 133	21
46		Cells 93/94 → LBC 142	21
47		Cells 90/91 → LBC 132	21
48		Cells 95/96 → LBC 143	21
49		Cell 96 − (pack −) → LBC 135	21
50		Cells 87/88 → LBC 138	21
51		Cells 75/76 → LBC 123	21
52		Cells 86/87 → LBC 130	21
53		Cells 78/79 → LBC 105	21
54		Cells 85/86 → LBC 137	21
55		Cells 81/82 → LBC 127	21
56		Cells 84/85 → LBC 108	21

Fig. 24 Module stack connectors: LB8 (left front) and LB7 (right front)

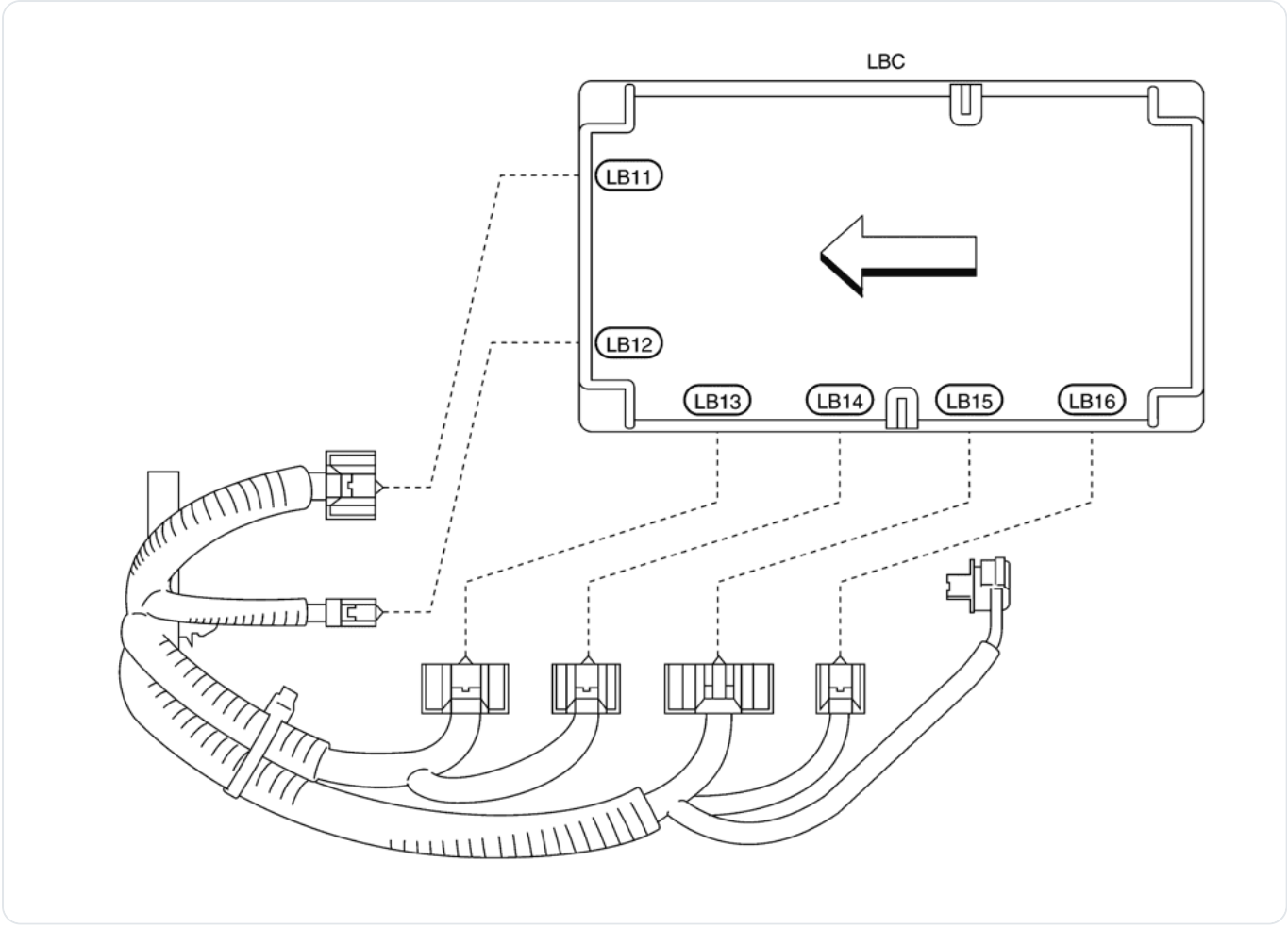


Fig. 25 LBC connector locations LB11–LB16 and harness

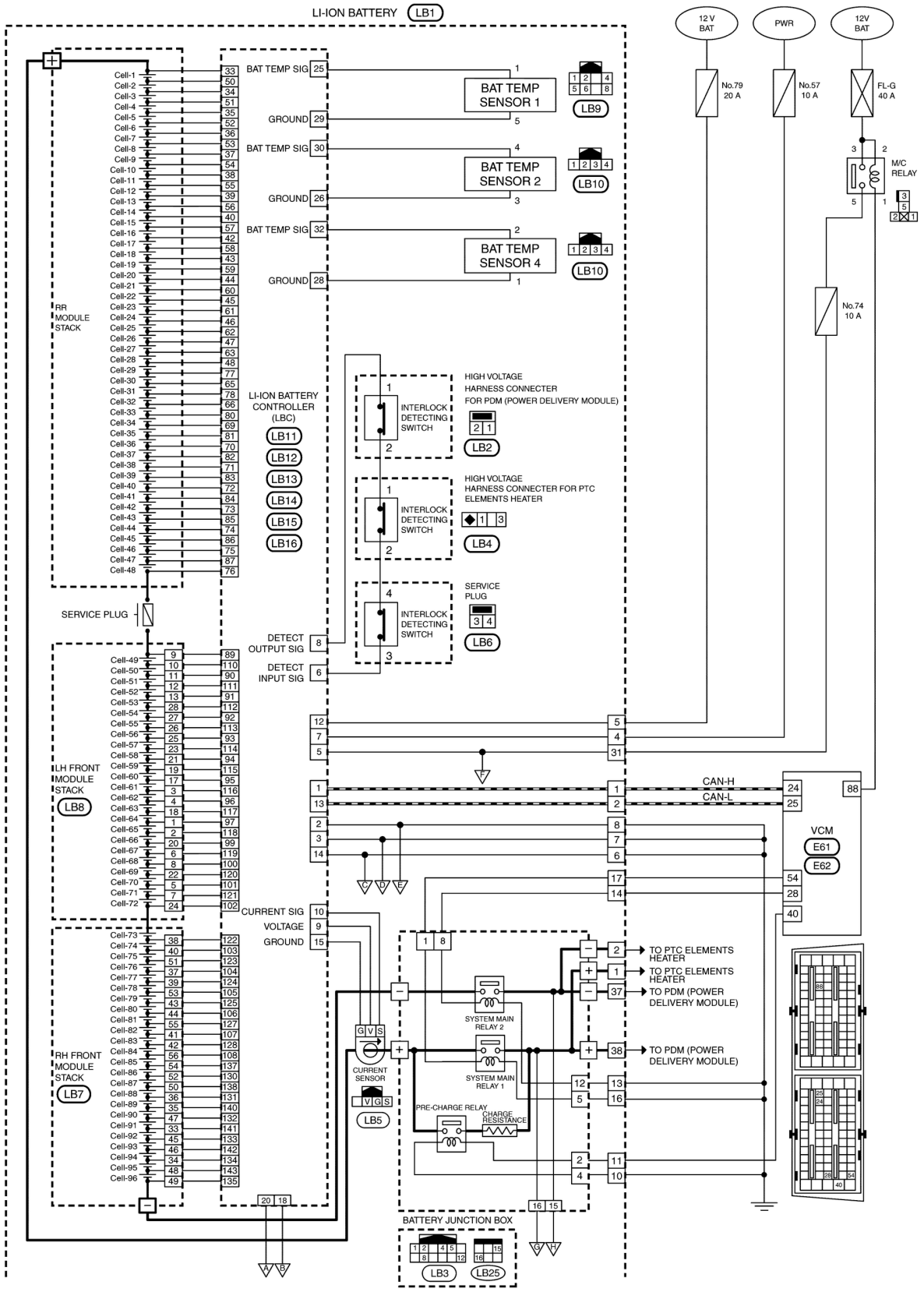


Fig. 26 Nissan Leaf AZE0 battery wiring — original factory drawing

> **NOTE**
> The diagrams are for the AZE0 BMS; other versions differ slightly.

8.3 Orion 2

Supported by the controller. Detailed documentation will be added in future revisions.

9 Current Sensor

The current sensor can be placed anywhere inside the battery before the power relays.

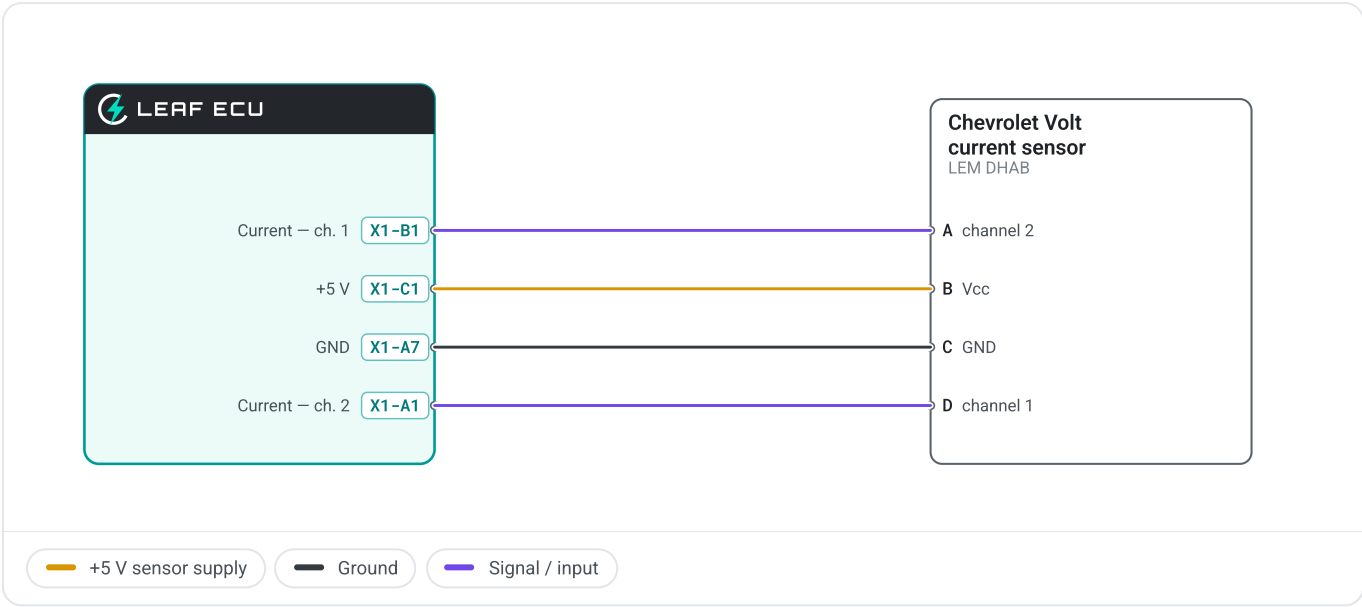
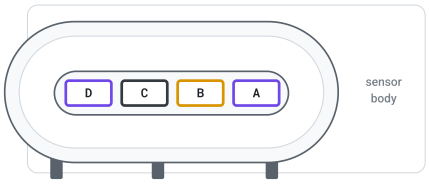


Fig. 27 Chevrolet Volt current sensor wiring



Front view into the sensor connector (as in the LEM drawing)

PIN	WIRE	FUNCTION	DIAGRAMS
A		Channel 2 → ECU X1-B1 (SIG 1)	27
B		Vcc, +5 V ← ECU X1-C1	27
C		Gnd, ground ← ECU X1-A7	27
D		Channel 1 → ECU X1-A1 (SIG 2)	27

Fig. 28 Chevrolet Volt current sensor connector (LEM DHAB)

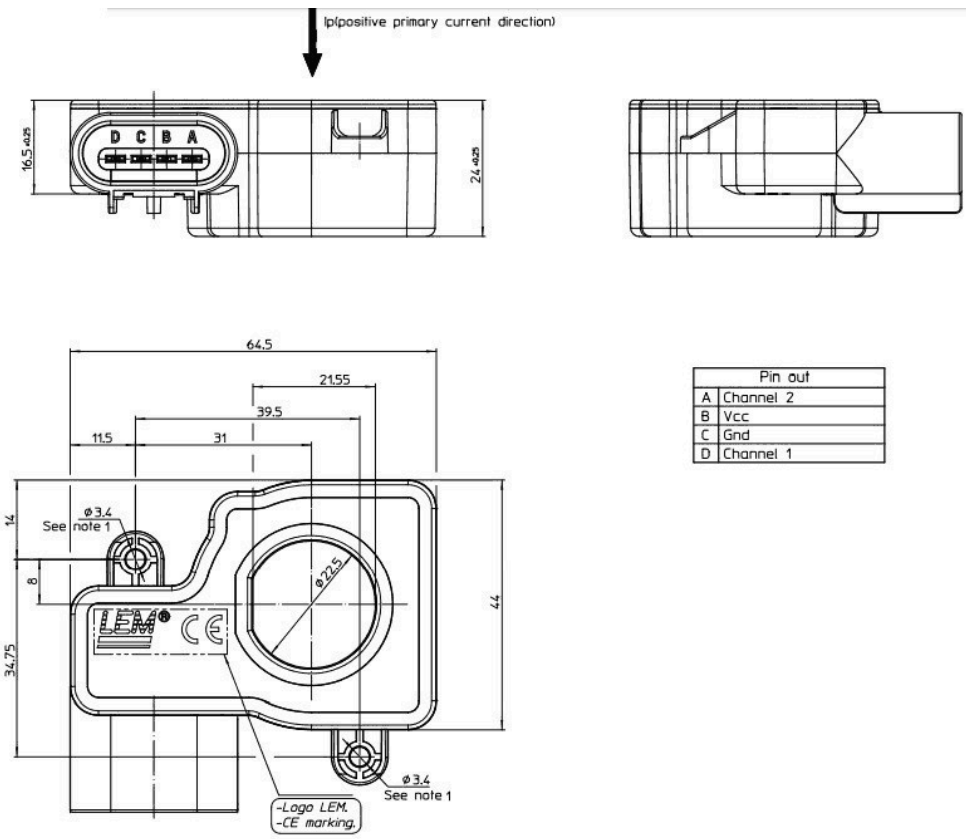


Fig. 29 LEM DHAB current sensor: drawing, pinout and current direction



Fig. 30 Current sensor connector

10 Cooling System

A liquid cooling system is required for the powertrain to work properly.

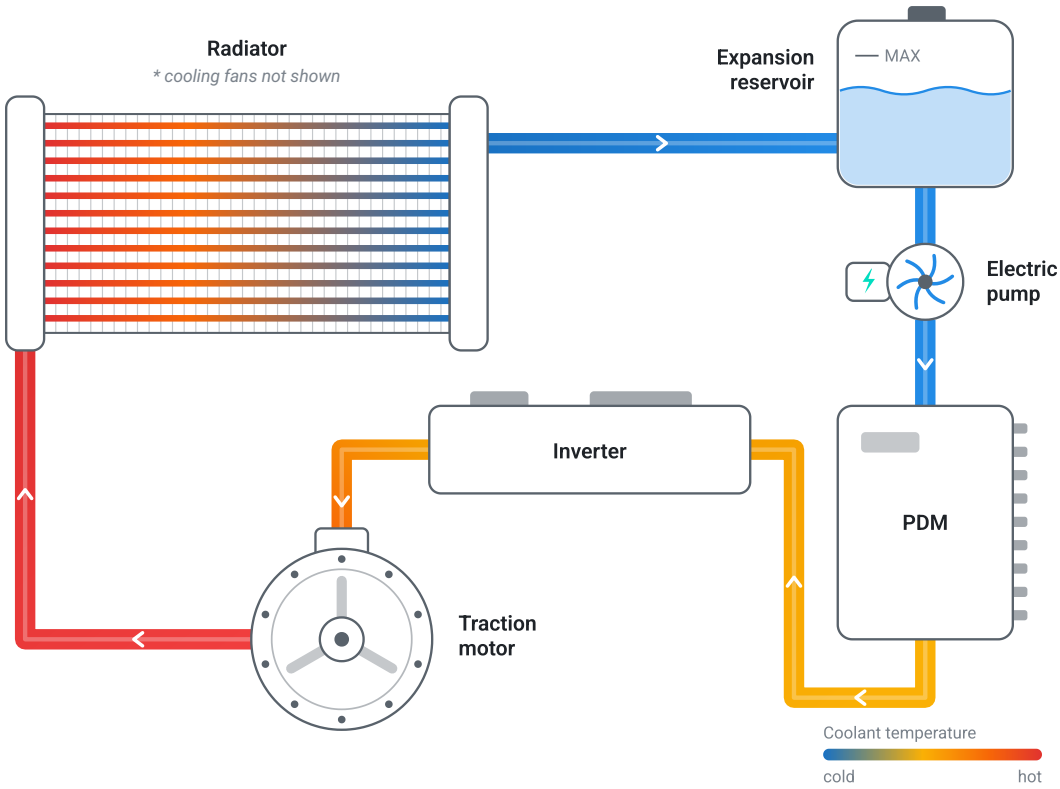


Fig. 31 Liquid cooling loop: radiator → reservoir → pump → PDM → inverter → motor

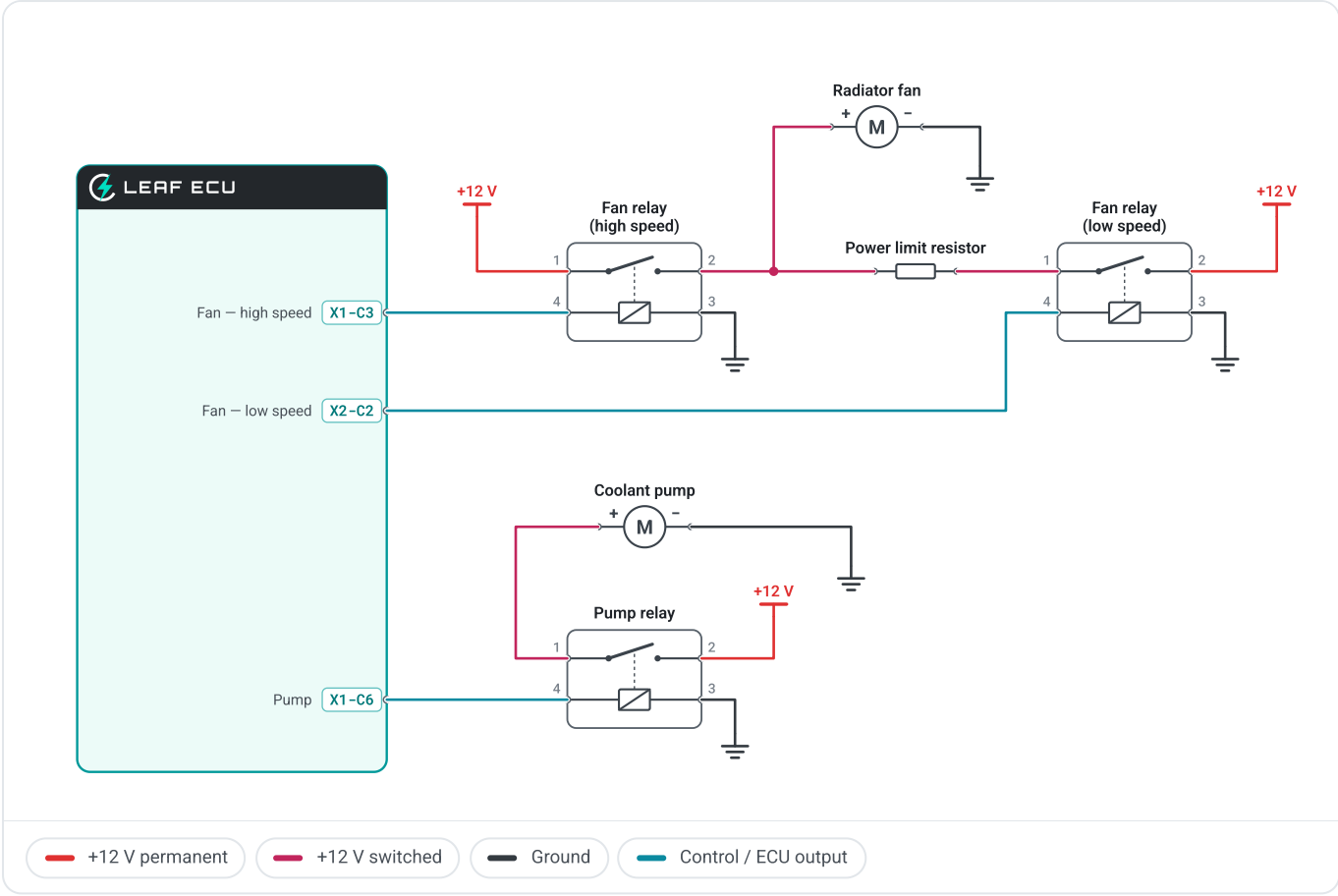


Fig. 32 Radiator fan (two speeds) and coolant pump control

Pump and fan switch-on temperatures are set in the configurator. The controller can run the fan at two speeds:

- full;
- reduced — through a power-limiting resistor.

The fan can also be driven by an external PWM controller (without the resistor).

11 Brake Light (Regeneration)

The controller can switch on the brake light during regenerative braking. To do this, install a relay in parallel with the brake pedal switch.

Feed the brake-light signal to pin X2-B8 so the controller can see the brake pedal being pressed.

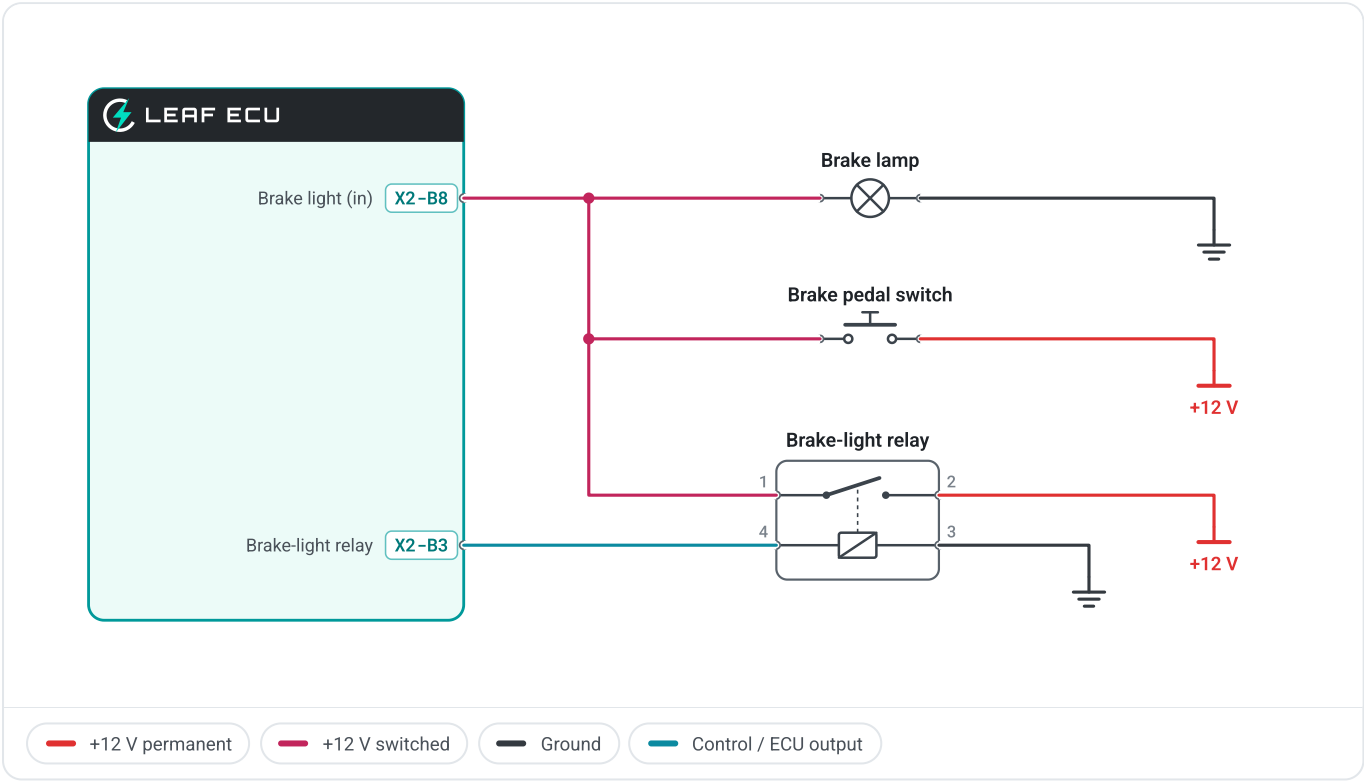


Fig. 33 Brake light during regeneration: relay in parallel with the brake pedal switch

12 CHAdeMO

If the PDM has a CHAdeMO port, its internal power contactors (CHAdeMO Power Relay + and −) can be used by running separate wires from them. If the PDM has no CHAdeMO port, two additional power contactors must be installed.

The Enable Relay is installed separately regardless of the PDM version. It does not need to be high-power — only strong enough to switch the power contactor coils.

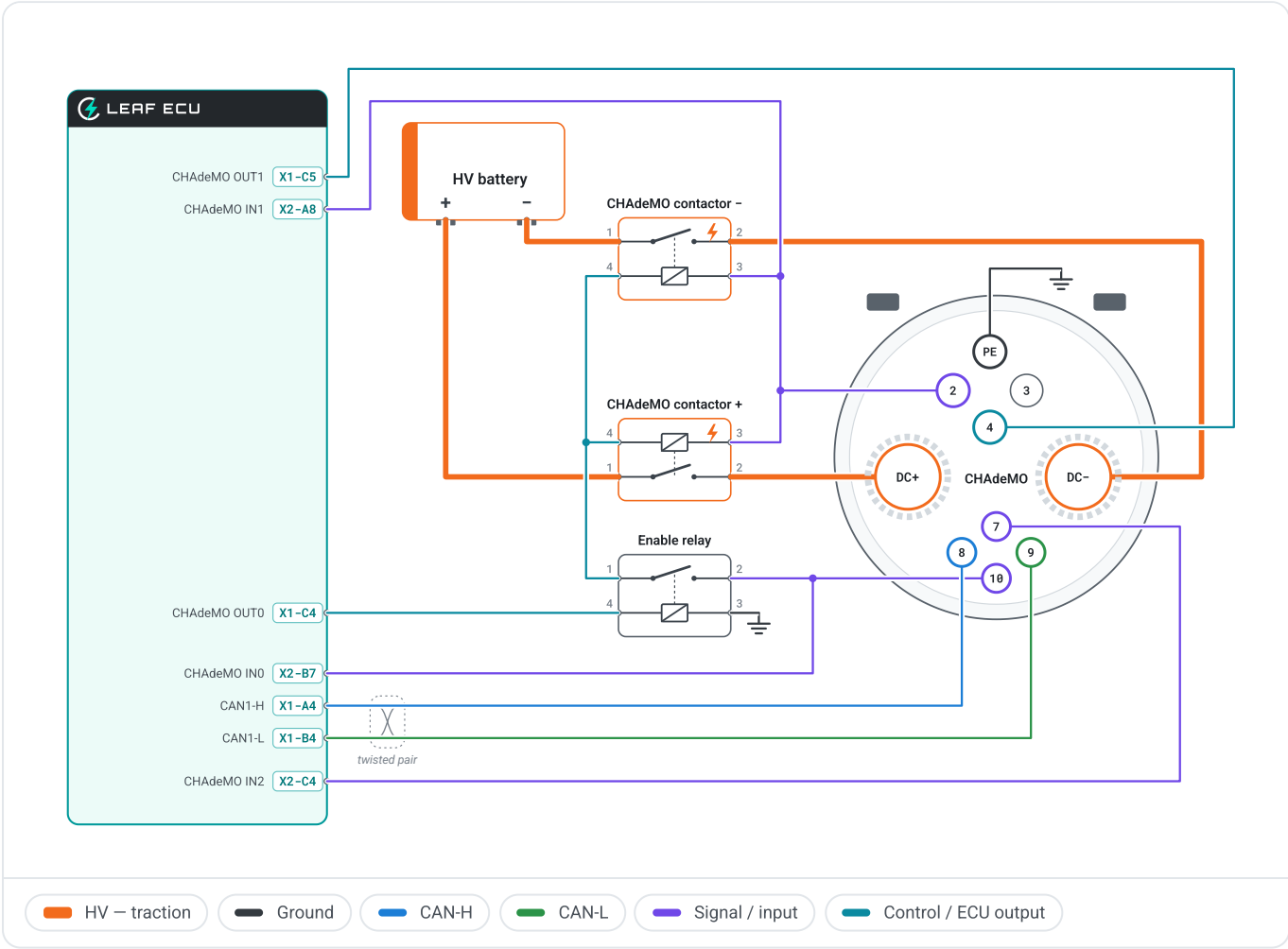
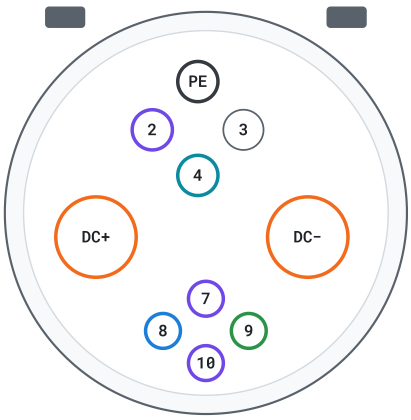


Fig. 34 CHAdeMO inlet wiring: power contactors, enable relay and charge sequence signals



Front view of the vehicle inlet

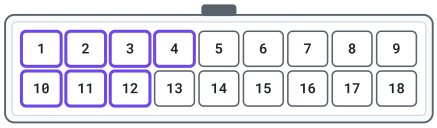
PIN	WIRE	FUNCTION
2		Charge sequence signal 1 → ECU X2-A8
3		Not used
4		Vehicle charge permission ← ECU X1-C5
7		Proximity detection → ECU X2-C4
8		CAN-H → ECU X1-A4
9		CAN-L → ECU X1-B4
10		Charge sequence signal 2 → ECU X2-B7
DC-		DC− → CHAdeMO contactor −
DC+		DC+ → CHAdeMO contactor +
PE		Protective earth (FG) → ground

Fig. 35 CHAdeMO inlet

13 Display

The display controller sits on CAN0 Powertrain and reads the warning-lamp states through its own inputs.

Display controller — connector X1 (signal-lamp inputs)



PIN	WIRE	FUNCTION
1		Right turn signal (IN1)
10		Left turn signal (IN2)
2		Parking lights (IN3)
11		Low beam (IN4)
3		High beam (IN5)
12		Parking brake (IN6)
4		Seat belt (IN7)

Display controller — connector X2 (Nextion display)



PIN	WIRE	FUNCTION
1		TX — data out to the Nextion display
3		RX — data in from the Nextion display
2		+5 V — display supply output
4		Display ground (GND)

Display controller — connector X3 (interface)



PIN	WIRE	FUNCTION
3		CAN-H — Powertrain bus (CAN0)
7		CAN-L — Powertrain bus (CAN0)
4		+12 V — controller supply
8		Ground (GND)

Fig. 36 Display controller connectors: X1 — warning-lamp inputs, X2 — display, X3 — interface

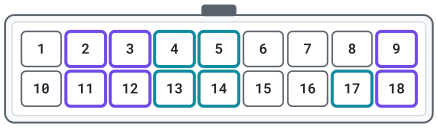
14 Selector

To use the Leaf selector or separate momentary buttons with position backlighting, an additional selector controller is required. Like the display, it connects to the main Powertrain CAN bus; its +12 V comes from the relay that powers the ECU.

Pin `X1-17` of the selector controller is the reverse-light signal (in both wiring options).

> **NOTE**
> Pin numbering on the selector connector itself may be mirrored. Wire according to the diagrams below — they show the connectors from the wire side.

Selector controller — connector X1 (selector / buttons)



Wire-side view

PIN	WIRE	FUNCTION	DIAGRAMS
9		Leaf selector: → selector pins 1, 7	38
18		Leaf selector: → selector pins 6, 12	38
2		Leaf selector: → selector pin 8 · Buttons: button P	38 40
11		Leaf selector: → selector pin 9 · Buttons: button R	38 40
3		Leaf selector: → selector pin 3 · Buttons: button N	38 40
12		Leaf selector: → selector pin 4 · Buttons: button D	38 40
4		Buttons: P backlight (LED anode)	40
13		Buttons: R backlight (LED anode)	40
5		Buttons: N backlight (LED anode)	40
14		Buttons: D backlight (LED anode)	40
17		Reverse light — relay coil (both configurations)	38

Selector controller — connector X3 (interface)



Wire-side view

PIN	WIRE	FUNCTION	DIAGRAMS
4		+12 V — from the relay that powers the ECU	38 40
8		Ground (GND)	38 40
3		CAN-H — Powertrain bus (CAN0)	38 40
7		CAN-L — Powertrain bus (CAN0)	38 40

Fig. 37 Selector controller connectors

14.1 Leaf Selector

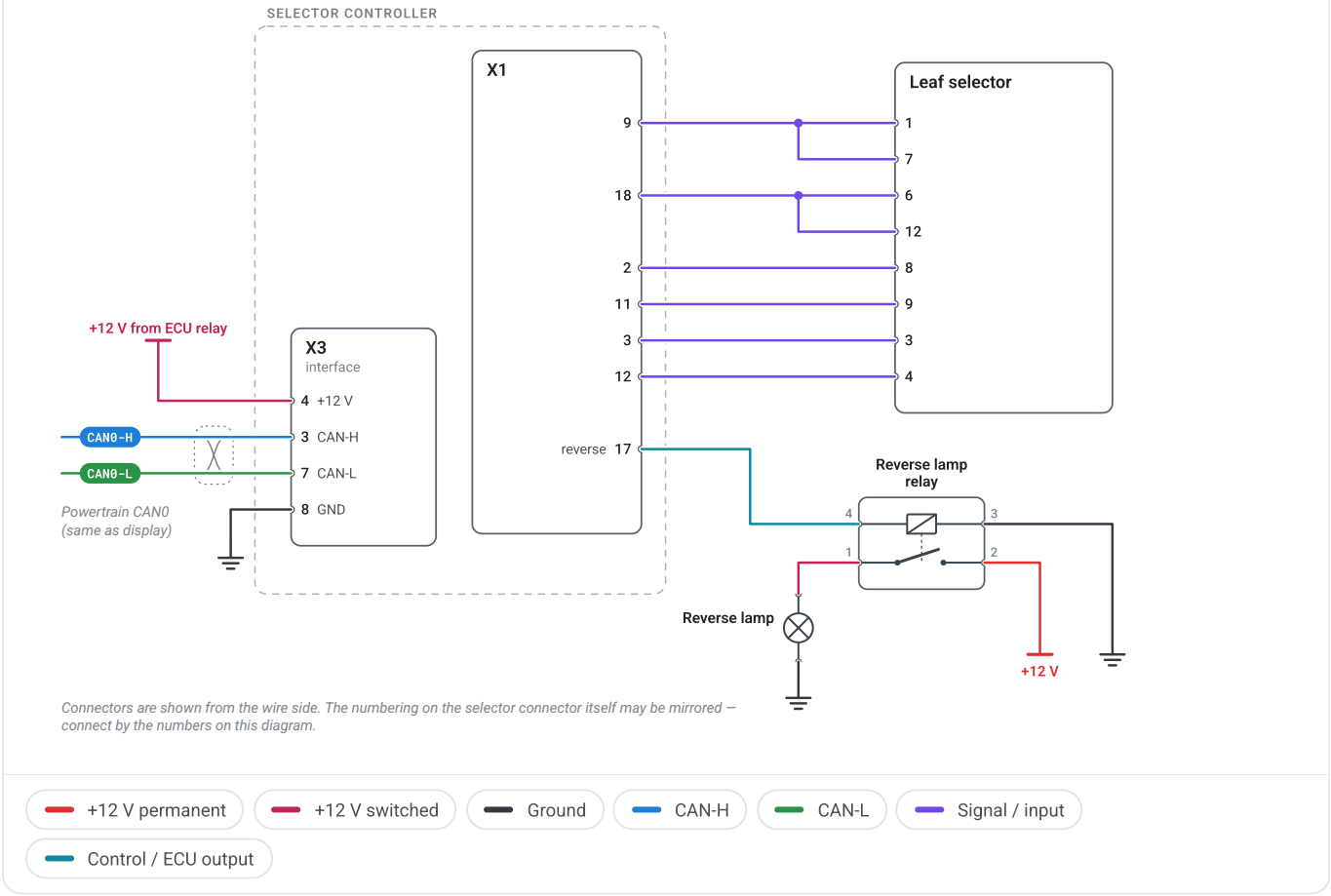
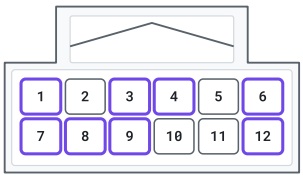


Fig. 38 Selector controller with the Nissan Leaf selector and the reverse-lamp relay



H.S. — harness (wire) side view

PIN	WIRE	FUNCTION	DIAGRAMS
1		To selector controller X1-9 (together with pin 7)	38
7		To selector controller X1-9 (together with pin 1)	38
6		To selector controller X1-18 (together with pin 12)	38
12		To selector controller X1-18 (together with pin 6)	38
8		To selector controller X1-2	38
9		To selector controller X1-11	38
3		To selector controller X1-3	38
4		To selector controller X1-12	38

Fig. 39 Nissan Leaf selector — 12-way connector

14.2 Momentary Buttons

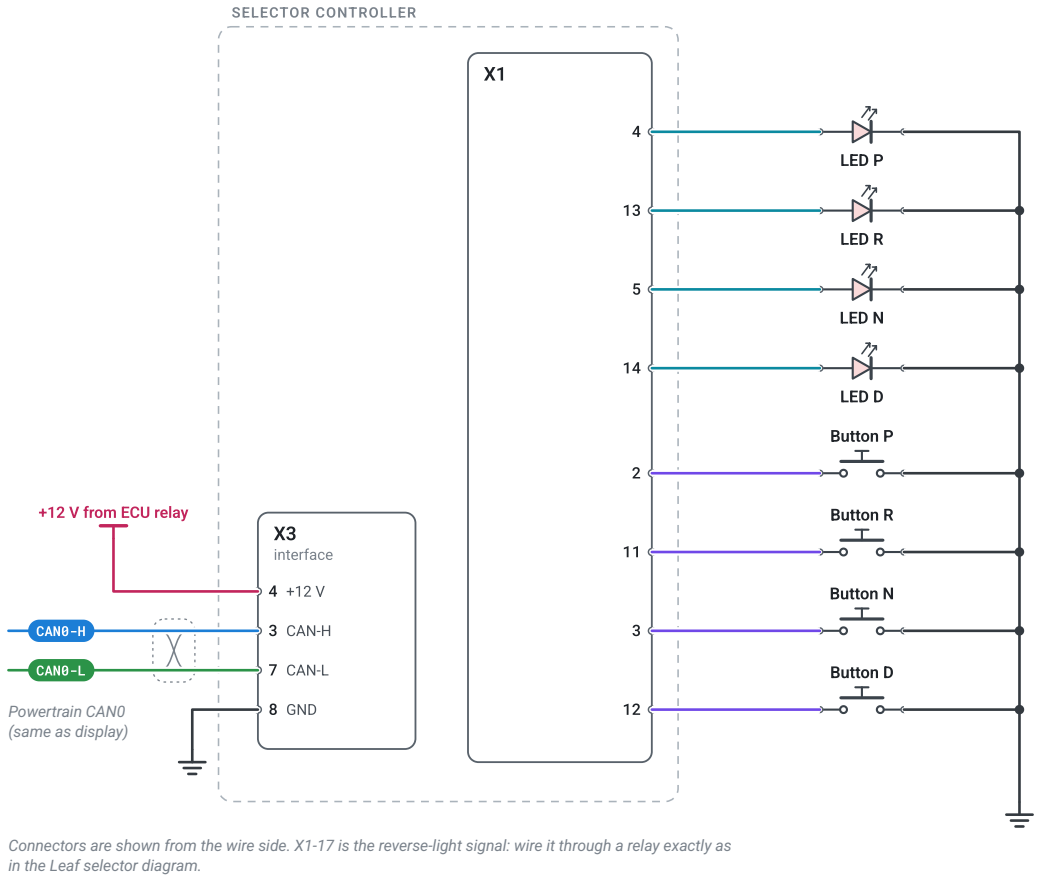
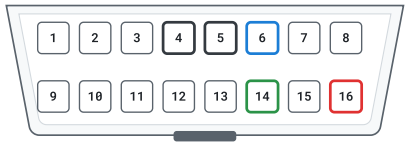


Fig. 40 Selector controller: momentary P/R/N/D buttons with backlight LEDs

15 OBD2 Diagnostics

15.1 OBD2 Connector

An OBD2 socket must be installed for controller setup and diagnostics. Use an ELM327 v1.5 adapter (Bluetooth, Wi-Fi, USB) or an OBDLink.



Front view, from the scan-tool side

PIN	WIRE	FUNCTION	DIAGRAMS
4		Chassis ground	
5		Signal ground (GND)	2 3
6		CAN-H (Powertrain, CAN0)	2 3
14		CAN-L (Powertrain, CAN0)	2 3
16		+12 V permanent	2 3

Fig. 41 OBD2 socket (vehicle side)

15.2 Software

- **DDT4ALL** — controller configuration and parameter monitoring (Windows, Linux).
- **CanZE** — Android app: cell map and battery temperature.
- Online diagnostics — the E-Engines web app for ELM327 in the browser.

16 Initial Setup

Once all wiring is connected and carefully checked (make sure no connector is pinned mirrored), proceed with the setup.

16.1 Start-up Procedure

Step 1. Power checks (before connecting the controller to the car)

- ☐ Permanent +12 V on pin `X1-A8` relative to `X1-A7` (ground).
- ☐ +12 V from the ignition switch on pin `X1-C8`.
- ☐ Charge inlet without a gun: resistance between `X1-B7` and `X1-A7` is **4.7 kΩ** or **2.7 kΩ** (depending on inlet type).
- ☐ Charge inlet with the gun plugged in: resistance between `X1-B7` and `X1-A7` is **infinite (OL)**.

Step 2. Wake-up and sleep checks

- ☐ Connect the controller to the car.
- ☐ Turn the ignition on — power appears on `X1-B8`, the IGN relay closes.
- ☐ Turn the ignition off and wait 10 s — power on `X1-B8` disappears, the IGN relay opens.
- ☐ Check wake-up and sleep by plugging and unplugging the AC charging gun.

Step 3. Configuration with DDT4ALL

> **DANGER — HIGH VOLTAGE**
> Do not continue without configuring the controller — the motor may spin at full speed or not spin at all!

- ☐ Connect the computer via ELM327 and DDT4ALL.
- ☐ **INV** tab: motor and inverter temperatures, no errors.
- ☐ **PDM** tab: PDM module temperature.
- ☐ Selector: the position must change on the **INV** tab.
- ☐ Pedal: on the **INV** tab "throttle pedal position" must rise as you press the pedal.

Step 4. Power relay and battery polarity checks

1. Disconnect the traction battery from the powertrain (pull the service disconnect).
2. Connect a voltmeter to the battery output after the contactors.
3. Press Start — the voltmeter must show voltage of the correct polarity.
4. The controller enters protection mode and reports a precharge error — this is expected.
5. Restart the controller (let it go to sleep and switch it on again).
6. Repeat the test measuring resistance across the positive contactor with a multimeter:
   - before start — infinite;
   - at start — 30–50 Ω;
   - after one second — 0 Ω.

16.2 Accelerator Pedal Setup

The parameters are on the configurator's **INV** tab.

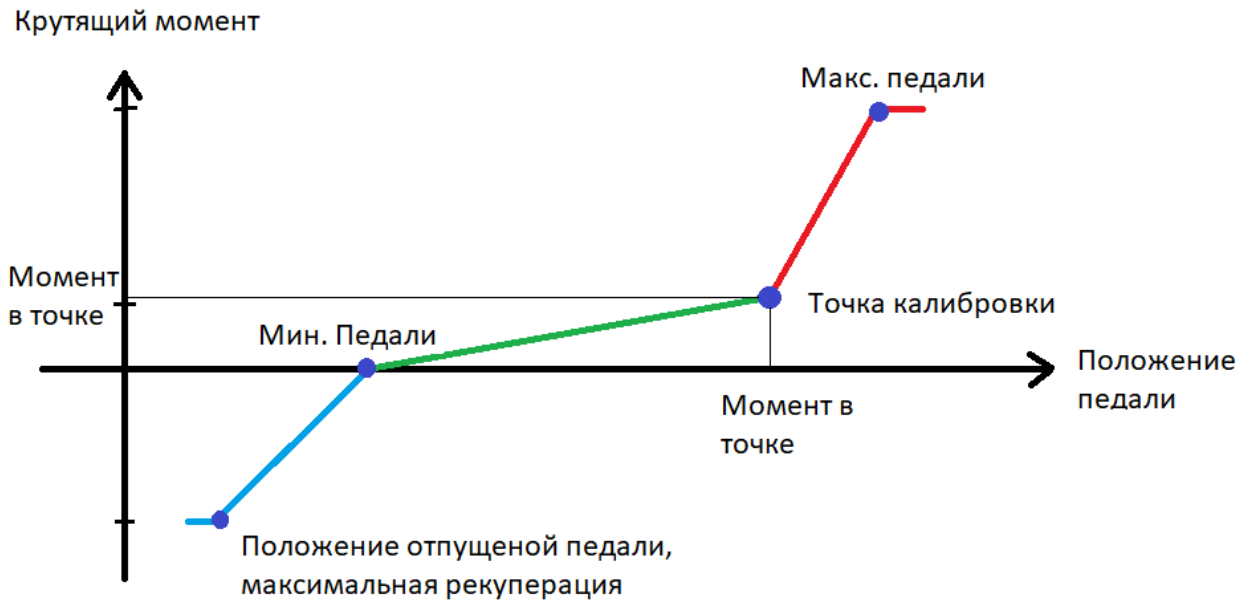


Fig. 42 Accelerator pedal parameters in DDT4ALL

PARAMETER	DESCRIPTION
Throttle min	"Throttle pedal position" value at which the motor starts to turn
Throttle max	"Throttle pedal position" value at which the motor turns at full power
Released pedal position	"Throttle pedal position" value with the pedal fully released
Torque at point	Percentage of maximum torque at the calibration point
Pedal position at point	"Throttle pedal position" value at the calibration point
Idle torque	Torque with D or R engaged and the pedal released (0 Nm = off)

16.3 Controller Configuration

After the pedal is set up, you must set **Maximum torque**, then open the **Setting** tab and check every parameter.

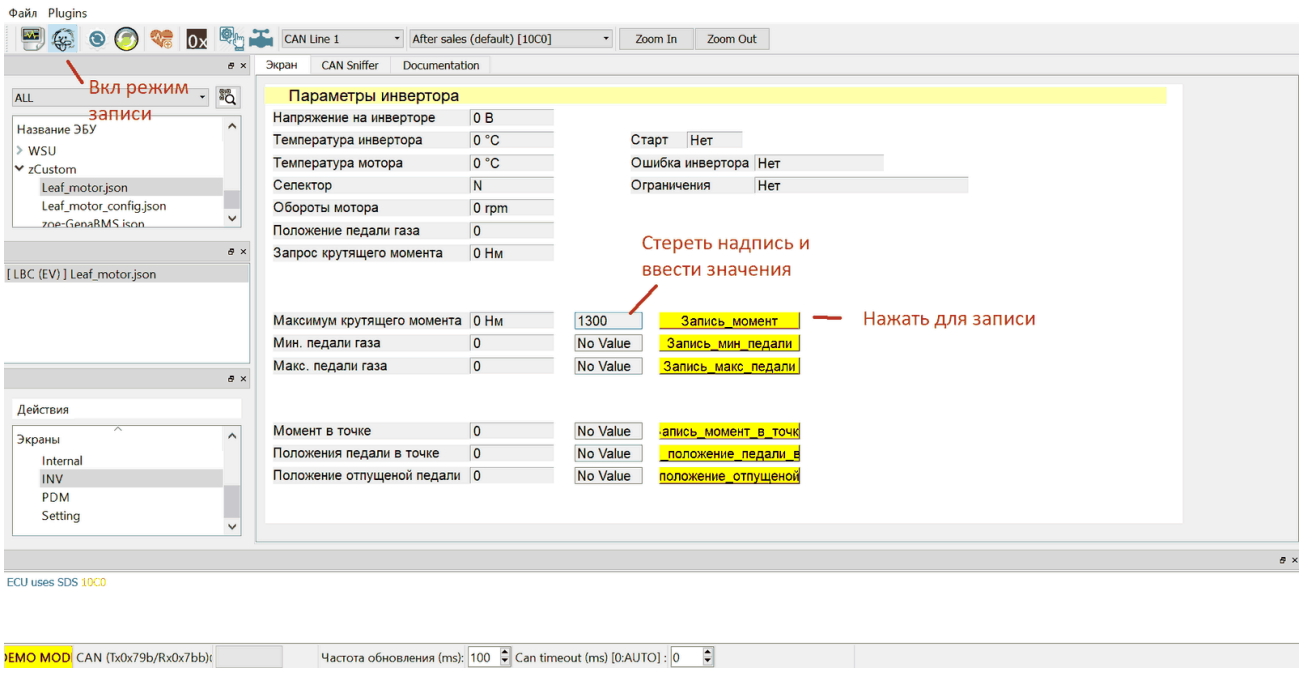


Fig. 43 Controller configuration parameters in DDT4ALL

PARAMETER	DESCRIPTION
Wheel circumference	In cm
Inverter type	Select exactly the inverter installed in the car
Selector type	"Buttons" for a toggle switch, "Leaf" for the separate selector board
Contactor activation voltage	In volts. The main contactor will not close if the inverter voltage during precharge is below this value — protection against a broken power wire or precharge resistor

Parameter refresh buttons: — refresh once; — cyclic refresh (the rate is set with "Refresh rate" at the bottom of the page).

17 FAQ / Troubleshooting

DDT4ALL won't connect to the ELM

- Check that the ELM is paired: two COM ports should appear in the adapter selection window.
- Your ELM may not be version 1.5, or the driver is wrong. Use a dual-board ELM and update the driver.

DDT4ALL shows no parameters / "NO DATA"

The ELM has no CAN communication with the controller. Possible causes:

- no power to the controller;
- CAN bus wired incorrectly (CAN-H and CAN-L swapped);
- controller fault.

All voltages from the BMS = 0 V

The controller does not see the BMS. Check the CAN bus wiring and the BMS power supply.

One cell reads much higher and its neighbour much lower

Most likely a poor balancing-wire contact with the cell. If it happens under load — a poor contact in the power busbars between cells. Check every connection in the battery.

The cell map is not shown in DDT4ALL

This is a quirk of the software — the map may not appear on the first attempt. Press page refresh (the green circle). With cyclic refresh you may need to increase the refresh interval.

The motor whines but does not turn

With the original gearbox, check that it is not locked: turn both driven wheels in the same direction. If they don't turn, rotate the parking-lock shaft counter-clockwise.

The motor turns only in reverse

This happens when a ZE0 inverter is installed but a different inverter type is selected in the controller settings.

The car briefly accelerates or regenerates by itself while driving

Seen with resistive pedals: over time the resistor tracks wear and give an unstable voltage. **Solution:** replace the pedal.

The traction battery contactors weld

1. Check the contactor wiring.
2. Check the precharge resistor value.
3. Check the battery activation voltage parameter (above 200 V) in the configurator — it protects against a failed relay or precharge resistor.

The DC-DC converter in the PDM burned out

A ground fault (especially when the traction battery negative is tied to the 12 V system negative) can destroy the PDM.

DANGER — HIGH VOLTAGE

Always check the insulation resistance of the traction battery to the chassis.

A Complete ECU Pinout

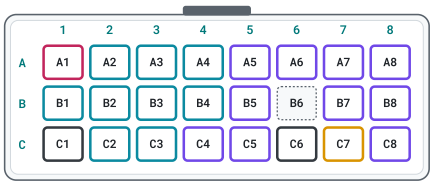
The pinout is compiled from the wiring diagrams in this manual. The "Diagrams" column lists the figures where the pin is used; hover a pin or a row to see it on the connector.



Row letters A–C and pin numbers 1–8 are moulded on the connector

PIN	WIRE	FUNCTION	DIAGRAMS
X1-A1		Current sensor — channel 2 (SIG 2)	27
X1-A2		Accelerator pedal — sensor 2 signal (not used by firmware yet)	13
X1-A3		CAN0 Powertrain — High, 500 kbps (inverter, PDM, OBD2, display, selector)	2 3
X1-A4		CAN1 — High: BMS (no CHAdeMO) or CHAdeMO	16 34
X1-A5		CAN2 — High: BMS when CAN1 is used by CHAdeMO	
X1-A6		CAN3 — High: expansion	
X1-A7		Ground	2 3 27
X1-A8		+12 V permanent from the battery (KL30)	2 3
X1-B1		Current sensor — channel 1 (SIG 1)	27
X1-B2		Accelerator pedal — primary sensor signal	13
X1-B3		CAN0 Powertrain — Low	2 3
X1-B4		CAN1 — Low	16 34
X1-B5		CAN2 — Low	
X1-B6		CAN3 — Low	
X1-B7		Charge-port PP line to the PDM (pin 29)	3
X1-B8		+12 V output to the IGN relay coil	3
X1-C1		+5 V output (pedal, sensors)	13 27
X1-C2		Ground (pedal, sensors)	13
X1-C3		Radiator fan relay — high speed	32
X1-C4		CHAdeMO output 0 — enable relay	34
X1-C5		CHAdeMO output 1 — charge permission (inlet pin 4)	34
X1-C6		Coolant pump relay	32
X1-C7		PP (proximity) from the charge inlet	3
X1-C8		+12 V from the ignition switch (KL15)	3

Fig. 44 ECU — connector X1 (black)



Row letters A–C and pin numbers 1–8 are moulded on the connector

PIN	WIRE	FUNCTION	DIAGRAMS
X2-A1		+12 V after the ignition relay	
X2-A2		Brake vacuum pump relay	10
X2-A3		Main contactor HV− coil	11
X2-A4		Main contactor HV+ coil	11
X2-A5		Start button input (+12 V = pressed)	9
X2-A6		Brake booster vacuum sensor signal	10
X2-A7		Analog input (via 10 kΩ) — reserved	
X2-A8		CHAdeMO input 1 — charge sequence 1 (pin 2)	34
X2-B1		Status LED / IGN (D28)	
X2-B2		Power output via MOSFET — reserved	
X2-B3		Brake-light relay (regeneration)	33
X2-B4		Precharge relay coil	11
X2-B5		Drive button input (+12 V = on)	9
X2-B6		Not connected (reserved)	
X2-B7		CHAdeMO input 0 — charge sequence 2 (pin 10)	34
X2-B8		Brake-light sense — brake pedal pressed	33
X2-C1		Ground	
X2-C2		Radiator fan relay — low speed	32
X2-C3		Brake-light relay (duplicate of X2-B3)	
X2-C4		CHAdeMO input 2 — proximity (pin 7) / AC gun	34
X2-C5		Reverse button input (+12 V = on)	9
X2-C6		Ground	16
X2-C7		+5 V output (sensors)	10 16
X2-C8		Digital input (optocoupler) — reserved	

Fig. 45 ECU — connector X2 (grey)

OBD2, display and selector connector pinouts are in their own sections: 15.1, 13, 14.

B CAN Buses

BUS	SPEED	INTERFACE	ECU PINS (H / L)	PURPOSE
CAN0 Powertrain	500 kbps	Hardware CAN0	X1-A3 / X1-B3	Inverter, PDM, OBD2, display, selector, CCS
CAN1	125–500 kbps	Hardware CAN1	X1-A4 / X1-B4	BMS (via NUP2105 protection) or CHAdeMO
CAN2	125–500 kbps	MCP2515 #1	X1-A5 / X1-B5	BMS when CAN1 is used by CHAdeMO; extra BMS modules
CAN3	500 kbps	MCP2515 #2	X1-A6 / X1-B6	Expansion

NOTE

Older diagrams numbered the buses from one ("CAN 1" = CAN0 Powertrain, "CAN 2" = CAN1). This manual uses the firmware names; go by the pin numbers.

Revision History

REVISION	CHANGES
2.0	New format: interactive web edition and PDF in Ukrainian and English. All diagrams redrawn in one style, connector views with pinouts added. Figure captions corrected; the ECU pinout is compiled from the wiring diagrams.
1.13	Translated to English. Improved structure: contents, numbered sections, figure captions, full pinout tables.
1.12	Added information on the selector and reverse light.
1.11	Added the selector CAN diagram.
1.10	Added FAQ and start-up procedure. Added the selector wiring diagram.
1.9	Fixed an error in the fan diagram. Added more information about IGN KEY.
1.8	Added the Volt BMS pinout.
1.7	Added Leaf BMS information. Fixed an error in the 12 V supply diagram with the charge-inlet resistor. Fixed an error in the current sensor diagram (pin A7 X1 → C7 X1). Added revision history.

ELECTRIC ENGINES

www.electricengines.com.ua

Interactive version with circuit highlighting:

https://diag.electricengines.com.ua/manual/en/

© E-Engines. Diagrams and text for installing the Leaf ECU controller.